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CHAPTER1

INTRODUCTION TO STUDENT INFORMATION MANAGEMENT SYSTEM

A student information management system will be basically a web application system that will run on local server for security issues. It will contain necessary information of students and teachers and courses. It will be used for education establishment by managing student data. It will provide capabilities for calculating students' evaluation marks, notifying them about course related events, generating student grade sheet and managing student related data needed in an institution. Not only that, using the system teachers can provide course materials, assignments to students as teachers and students both can use the system.

1.1 PURPOSE

The document is about the Software Requirements Specification (SRS) for the Student Information Management System. The purpose of this document is to give a detailed description of the requirements for the Student Information Management System. This software requirements specification document enlists all necessary requirements that are required for the project development. To derive the requirements we need to have clear and thorough understanding of the products to be developed. This has been prepared after detailed communications with the project team and stakeholders.

The document includes a set of use cases that describe interactions the users will have with the software. It will also explain the system constraints, interface and interactions between the software and the users.

1.2 OVERVIEW

- The remainder of this document includes eight chapters and appendices.
- The second and third chapters introduce different types of stakeholders and their interaction to the system. The chapters also provide the requirements specification in detail and provide a description of the different system interfaces.
- The fourth one provides an overview of the system functionality and system interaction with other systems based on the scenario of the Student Information Management System.
- The fifth and sixth chapters show the interaction of data within the system using various functionalities. Different specification techniques are used in order to specify the requirements more precisely for different audiences.
- The seventh chapter describes the behavior of the software.
- Conclusion is in the eighth chapter.
- The Appendices at the end of the document include the necessary tools we have needed to develop the SRS of the Student Information Management System

1.3 CONCLUSION

In this document we have discussed our project overview including project purpose. In next chapter the inception of this project will be described.

CHAPTER 2

INCEPTION OF STUDENT INFORMATION MANAGEMENT SYSTEM

2.1 INTRODUCTION

Inception is the beginning phase of requirements engineering. This phase helps to get orientation to make a first draft about project planning.

The inception phase is not responsible to describe the requirements completely and in detail, to restrict the problem field and also not to develop solutions. It defines how does this project (Student Information Management System**)** get started and what is the scope and nature of the problem to be solved. The goal of the inception phase is to identify concurrence needs and conflict requirements among the stakeholders of this project. To establish the groundwork we have worked with the following related to the inception phases:

- Planning meeting
- Identifying Stakeholders
- Recognizing multiple viewpoints
- Working towards collaboration factors
- Our questions to the stakeholders

2.2 PLANNING MEETING

At an early stage in the project, several stakeholders and subject matter experts should be convened to discuss the project and make the product plan. We should choose stakeholders based on the nature and complexity of the project and its product deliverable. Depending on the size of the project and its complexity, the meeting may take several days or weeks.

To make the project successful, we have made several times group discussion:

➢ Date: 13 September, 2015

Place: IIT

Subject Matter: Identifying stakeholders

Group Members:

- Farina Faiz (BSSE-0406)
- Md.Arafat Zaman Anik (BSSE-0410)
- Afrina Khatun (BSSE-0411)
- Md. Rakibul Islam (BSSE-0420)
- Mamun Khan (BSSE-0429)
- Zarif Masud (BSSE-0430)
- Kamrun Nahar Neela (BSSE-0431)
- Syed Ali Asif (BSSE-0432)

➢ Date: 4 October, 2015

Place: IIT

Subject Matter: Collecting requirements from stakeholders

Group Members:

- Above all members

➢ Date: 6 October, 2015

Place: IIT

Subject Matter: Discussion on requirements

Group Members:

- Above all members

➢ Date: 8 October, 2015

Place: IIT

Subject Matter: Discussion on report writing about “Inception” phase

Group Members:

- Above all members

2.3 Identifying Stakeholders

Stakeholder refers to any person or group who can affect or is affected by the system directly or indirectly. Stakeholders include end-users who interact with the system and everyone else in an organization that may be affected by its installation. Stakeholder identification is the process used to identify all stakeholders for a project. As we are going to develop student information management system we will hand it over to IIT for now that is why only the people of IIT (teachers, students and staff) will be permitted to use it. So the number of our stakeholders is small. Although the user number is small, they will use the system in different purpose and so have different types of requirements and viewpoints. It is important to understand that not all stakeholders will have the same influence or effect on a project, nor will they be affected in the same manner. It should be done in a methodical and logical way to ensure that stakeholders are not easily omitted. The following questions help us to identify stakeholders:

- Who uses the system?
- Who is affected by the outputs of the project?
- Who evaluates/approves system?
- Who maintains the system?
- Who has knowledge (specialist) about the system?
- Whose work will affect our project? (During the project and also once the project is completed).

2.4 Our Questions to the Stakeholders

We set our questions in a way so that they could give their opinions and requirements for the system. The questions are mentioned at the last part of SRS. Our questions also focus measurable benefits and successful implementation of the project.

2.5 Recognizing Multiple Viewpoints

Admin viewpoint

- The user interface should be designed in a way so that it looks understandable to the non technical persons
- The database should be secure
- Individual accounts for teachers and students must be created and their information must be uploaded
- Easy access
- Easy maintenance such as inserting, updating, deleting

Staff viewpoint

- Easy access
- User-friendliness
- File must be uploaded
- Provide all kinds of notifications

Teachers viewpoint

- Fully automated evaluation system
- Books and file must be uploaded
- Class routines must be shown
- Add assignments

Students viewpoint

- View evaluation marks continuously
- Get all kinds of notification
- View their assignments given by teachers

2.6 Working towards Collaboration

We have asked our stakeholders for their requirements and found out that each of them has their own requirements. Some of the requirements are common as well as conflicting. So we need to follow the steps given below to merge the requirements:

- Find out the common and conflicting requirements
- Divide the requirements into different categories
- Identify the specials requirements that the stakeholders have
- Identify the requirements according to the stakeholders priority points and prioritize them
- Take final decisions about the requirements

2.7 Conclusion

Inception phase helped us to achieve the concurrence among all stakeholders on the lifecycle objectives for the project as we recognized their multiple viewpoints. The inception phase is of significance primarily for new development efforts, in which there are significant requirements risks which must be addressed before the project can proceed. In next chapter we have discussed about finding requirements and user scenario.

CHAPTER3

ELICITATION OF STUDENT INFORMATION MANAGEMENT SYSTEM

Elicitation is the task that helps the customer to define what is required. To complete the elicitation step of Student Information Management System we faced many challenges like problems of the scope, problems of the volatility and problems of understanding. To help overcome these challenges, we have worked with the elicitation requirement activity in an organized and systematic manner.

3.1 ELICITING REQUIREMENTS OF STUDENT INFORMATION MANAGEMENT SYSTEM

Requirements elicitation of Student Information Management System combines elements of problem solving, elaboration, negotiation and specification. It requires the cooperation of a group of end users and developers to elicit requirements. To elicit requirements of Student Information Management System we completed following tasks:

- Collaborative Requirements Gathering
- Quality Function Development
- Usage scenario

3.2 COLLABORATIVE REQUIREMENTS GATHERING

The goal is to identify the problem, propose elements of the solution, negotiate different approaches, and specify a preliminary set of solution requirements in an atmosphere that is conducive to the accomplishment of the goal. To better understand the flow of events of Student Information Management System as they occur, we present a brief usage scenario that outlines the sequence of events that lead up to the requirements gathering meeting, occur during the meeting, and follow the meeting. We discuss with our stakeholders of Student Information Management System to come up with the deserved results. To gather the requirement, we have used a technique called use “4Ws and an H”. It refers to:

- Who
- What
- When
- Where
- How

There are three problems that are encountered as elicitation of Student Information Management System occurs:

1. PROBLEMS OF THE SCOPE: The boundary of the problem of Student Information Management System is well defined. We know what we are going to do and our limitation related to his project. to develop this system we will perform the following tasks:

- We will provide the access to the permitted users
- There will the options for inserting, deleting and updating the information of a user
- All data will be stored in database

2. PROBLEMS OF THE UNDERSTANDING: TO HAVE A CLEAR UNDERSTANDING ABOUT THE REASON OF DOING THIS AS A PROJECT, WE PUT OURSELVES IN FRONT OF SOME QUESTIONS SUCH AS:

- what is needed from the system
  - IIT’s students’ information and evaluation will be maintained by this system

- the capabilities and limitation of the computing environment
  - as now the IIT does not have such system the maintenance of students information is performed manually that lacks the security. It will make the whole work automated, smooth and authenticated.
- the domain of the problem
  - for now our system will be used by IIT after development but we have plan to make it available for other educational institutions.

3. PROBLEMS OF THE VOLATILITY: Requirements change overtime. This is due to dynamic environment. The problem of volatility is an important aspect.

4. REQUIREMENT ELICITATION THROUGH INTERVIEW: Interviews are often the best means to elicit qualitative information of Student Information Management System such as opinion, policies and narrative description of activities of the problem.

5. REQUIREMENT ELICITATION THROUGH QUESTIONING: Wide distribution questionnaires ensure that the respondent has great anonymity.

6. RECORD VIEW: A good volume of records and reports of Student Information Management System provide useful information about the existing system.

7. OBSERVATION: Observation provides first-hand information of Student Information Management System about how activities are carried out.

QUESTIONNAIRE OF THE PROJECT

The aim of the questionnaire is to collect information on the methods and practices for this project. Stakeholder's input is extremely important to identify gaps and desire data that will help in driving future tools and methods that may better support their activity. The sample questionnaire for the project is given below

QUESTIONNAIRES

- Why is it needed and why does the academic activity need to be made automated?
- What is the purpose and objectives of the project?
- What are the benefits?
- Who are the users?
- Will it be cost-effective?
- What are the benchmarks?
- How is the security ensured?
- How will it be useful to teachers and students?
- Is it feasible to add extra functionalities (like attendance module) in this app?
- Will it provide storage benefit?
- Can it be collaborated with social media (like email)?
- Is any Interaction module (like comment box) be available?

3.3 QUALITY FUNCTION DEVELOPMENT

Quality function deployment (QFD) is a quality management technique that translates the needs of the customer into technical requirements for software. QFD "concentrates on maximizing customer satisfaction from the software engineering process". With respect to our project the following requirements are identified by a QFD.

NORMAL REQUIREMENTS

- This app will contain students records and evaluation marks
- Teachers and students will have separate profiles

- Admin will create all users, their role and can delete user profile if necessary
- Users can edit their information
- Students can monitor their own evaluation marks continuously
- Teachers can view their class routine from dashboard
- Staffs will have accounts
- Database will be secured
- Easily maintainable
- Easily accessible
- Notice will be provided by staffs
- Notice about exams and assignments will be provided by teachers
- Help feature to explain what they are looking for
- Course information will be categorized
- The user interface will vary based on users

Expected Requirements

- The user interface of the system shall be easy to use
- Different evaluation and marking criteria will be available
- Attendance management module to manage attendance
- File uploading will be enabled
- Teachers and students will receive notifications

Exciting Requirements

- The system will send email to the user when a notification is created
- Teachers and students will receive live notification about academic and other events
- Dynamic notification bar for teachers and students
- Error message will be provided
- Comment box for questions and queries

3.4 Usage Scenario

Student Information Management System is a web application which will run in local servers due to security issues. This app will contain all the necessary information of teachers, students and courses of Institute of Information Technology (IIT). Users of this application can be categorized as following:

- Administrative Users (Admin)
- Staffs
- Teachers
- Students

Administrative users (Admin) will create accounts of teachers, students and staffs. By creating roles of the users, admin will distinguish among different users. Access levels, services and user interface will vary based on these roles. Admin will also store all the required information of teachers and students in order to complete the user's profile. Admin will also delete/inactive profiles if situation demands (for example, after a student completes his/her graduation program or if a teacher leaves the Institution).

Staffs will assign requisite courses to teachers and students. Hence when a teacher logs into his/her account he/she can get to know which courses are to take. Students will also get to know which courses are obligatory for them. Staffs can also provide notices by uploading files regarding holidays, vacations, exam schedules, routines and others. These notices will be comprised of subjects and descriptions. Event notification about these notices will be sent to intended receivers.

Teachers are one of the main users of this app. After logging in teachers can view the list of assigned courses to them. Every course will be maintained in distinct dashboards. In this dashboard, teacher has to enter different categories of marks on which evaluation will be done. If a course is selected, list of students and their rolls of this course will be displayed in the dashboard automatically. Teacher can create categories of marks such as final, mid-term, presentation, project, continuous evaluation and lab. Teacher can also create several sub-categories under each category. Teacher will enter these types of marks in the dashboard. Calculated percentage of these marks will be available in the dashboard. Teachers can set their own rules for calculating marks and can take as many as exams as they like. These rules include setting up different weights on different categories and consider selected categories for calculation. All of these will be stored in the dashboard. After teachers finish entering all the marks, final percentage of the marks will be available in the dashboard. After teachers press “finish” button, students can access their final result from the dashboard after logging in the app. Students will be notified of the result publication via notifications. This result can also be downloaded as pdf by the specified student. Teacher can edit all the information of the dashboard and can also upload requisite books, files & notes for student’s usage. Students can view and download these documents. Teachers can also give a notice about the next course exam in the dashboard and students of that course will receive immediate notification about the exam. If a new course is assigned to a teacher, he/she will be notified immediately via the app. Teacher can answer questions regarding the course asked by students in the specified comment box. Teacher can also manage attendance of the students from attendance management module of the app. This attendance sheet will only be enabled for students after teacher publishes the attendance. After attendance publication attendance marks will be added to the continuous evaluation of the student by rules defined by the teacher. Final report will be available to students after teacher publishes it.

Students can view the course marks after logging in the app. Students will also receive notifications from staffs and teachers. Students can express their queries to teacher about the course in the specified comment box for the course. Students can view their course attendance only after attendance publication made by course teacher. Attendance marks will also be added to the continuous evaluation for the students.

Therefore, every users of the app will have an account and a notification bar assigned for them. If new notification loads in the notification bar an email will be sent to his/her Gmail account to notify him/her about new notifications. This app will be designed to manage, store and publish results and other notices regarding holidays, vacations & exams. This app will make life easy for teachers, students and academic staffs by eliminating pen & paper and ensuring transparency between teachers and students.

CHAPTER 4

SCENARIO BASED MODELING OF STUDENT INFORMATION MANAGEMENT SYSTEM

This chapter describes the scenario based modeling of Student Information Management System.

4.1 INTRODUCTION

Scenario based modeling comprises of three diagrams:

- Usecase diagram
- Activity diagram
- Swimlane diagram

Based on the scenario described in previous chapter, we have designed usecase diagram, activity diagram and swimlane diagram of the project.

A usecase diagram is a graphic depiction of the interactions among the elements of a system. A usecase is a methodology used in system analysis to identify, clarify and organize system requirements and it contains some components.

- the actors, usually individuals involved with the system defined according to their roles
- the use cases, which specific roles are played by the actors within and around the system
- the relationships between and among the actors and the use cases

In student information management system the usecase diagram represents the whole process and activities of managing students' information.

The actors are to be identified from the scenario.

These actors are of two types:

- Primary Actor
- Secondary Actor

PRIMARY ACTOR: Primary actors interact directly to achieve required system function and derive the intended benefit from the system. They work directly and frequently with the software.

SECONDARY ACTOR: Secondary actors support the system so that primary actors can do their work. They either produce or consume information.

The identified actors of student information system are:

- Admin
- Staff
- Teacher
- Student

4.2 USECASE DIAGRAMS AND SCENARIO

In this section usecase diagram and scenario are described elaborately. Each usecase diagram contains its name, level number, primary and secondary actors.

The scenario of the first usecase diagram is provided in previous chapter as it represents the whole scenario of the student information system.

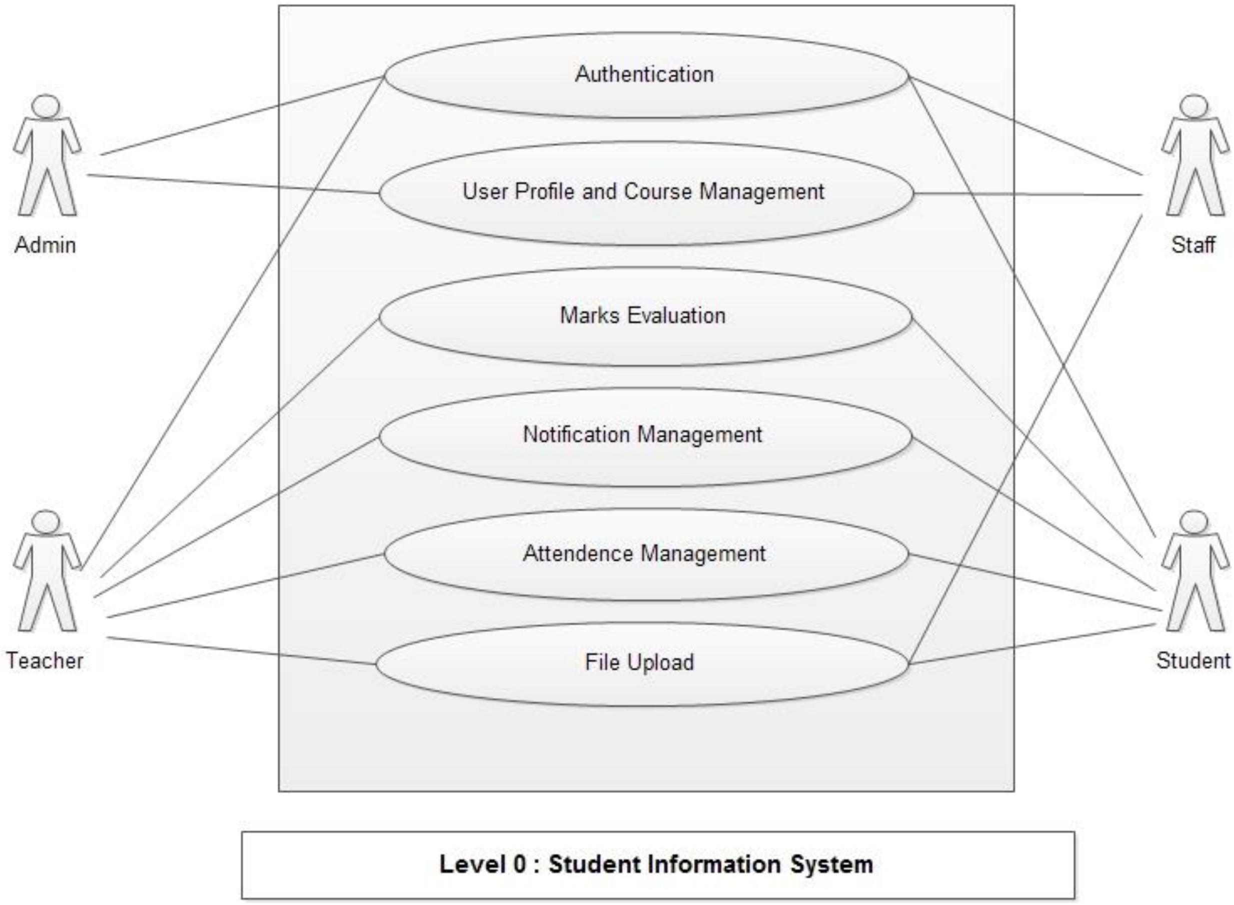

FIGURE 4.2.1: STUDENT INFORMATION MANAGEMENT SYSTEM (LEVEL 0)

Usecase Name: Student Information Management System

Usecase ID: 0

Primary Actor: Admin, Teacher, Staff

Secondary Actor: Student

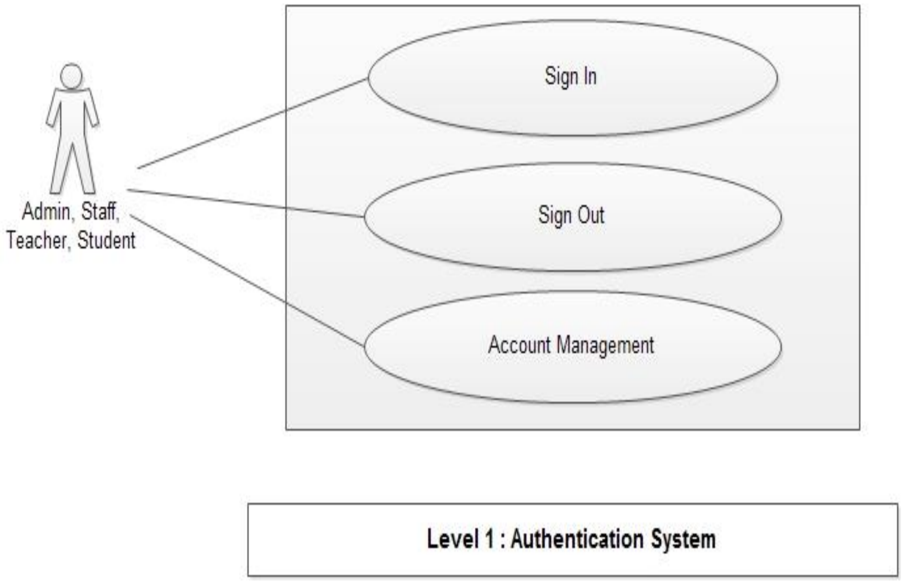

FIGURE 4.2.2: AUTHENTICATION SYSTEM (LEVEL 1)

Usecase Name: Authentication System

Usecase ID: 1

Primary Actor: Admin, Teacher, Staff, Student

Scenario for level 1: Through the authentication system the will login to the system by giving a particular id and password and the system verifying the id and password will provide access to the users. The users will logout from the system after performing their task in the system.

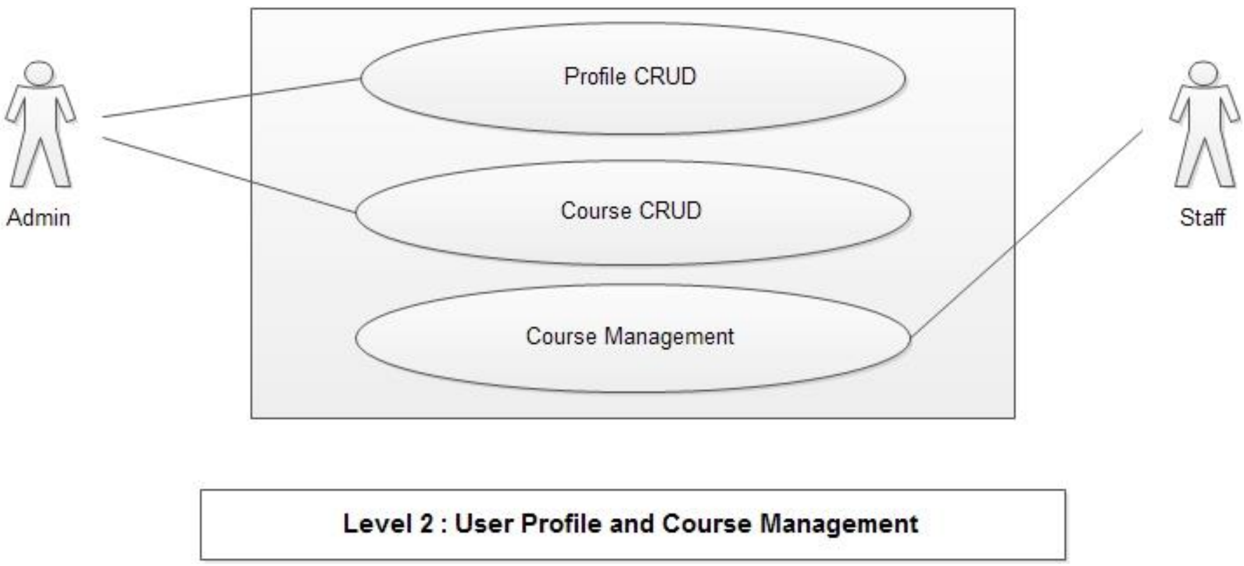

FIGURE 4.2.3: USER PROFILE AND COURSE MANAGEMENT (LEVEL 2)

Usecase Name: User Profile and Course Management

Usecase ID: 2

Primary Actor: Admin, Staff

Scenario for level 2: Administrative users (Admin) will create accounts of teachers, students and staffs. He/She will also delete/inactive profiles if situation demands. He and Staff will assign requisite courses to teachers and students.

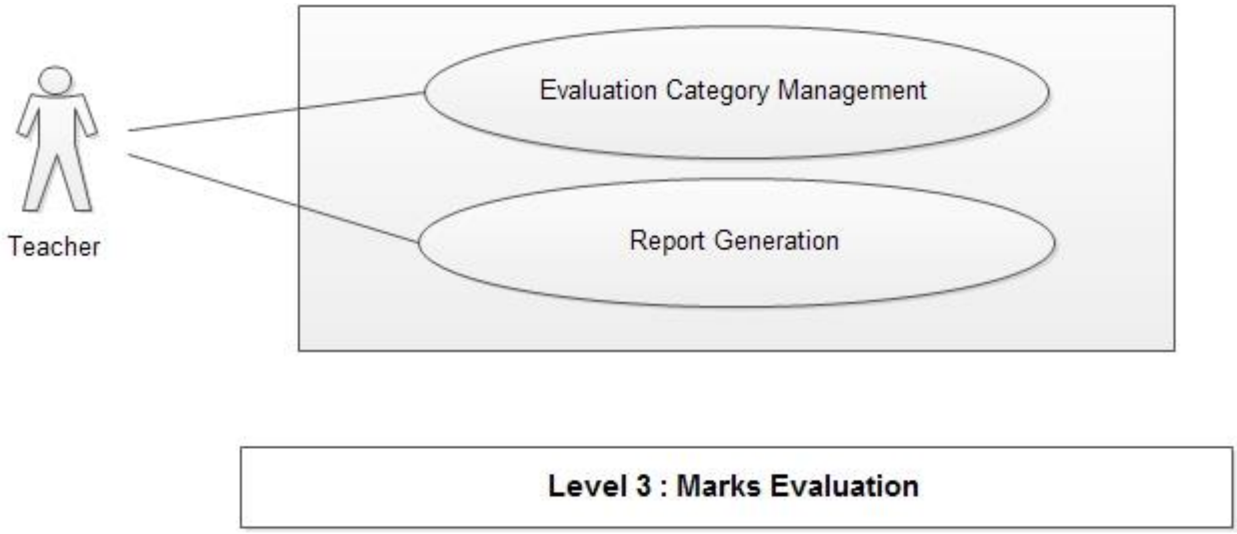

FIGURE 4.2.4: MARKS EVALUATION (LEVEL 3)

Usecase Name: Marks Evaluation

Usecase ID: 3

Primary Actor: Teacher

Scenario for level 3: After logging in teachers can view the list of assigned courses to them. Every course will be maintained in distinct dashboards. In this dashboard, teacher has to enter different categories of marks on which evaluation will be done. Final report will be available to students after teacher publishes it.

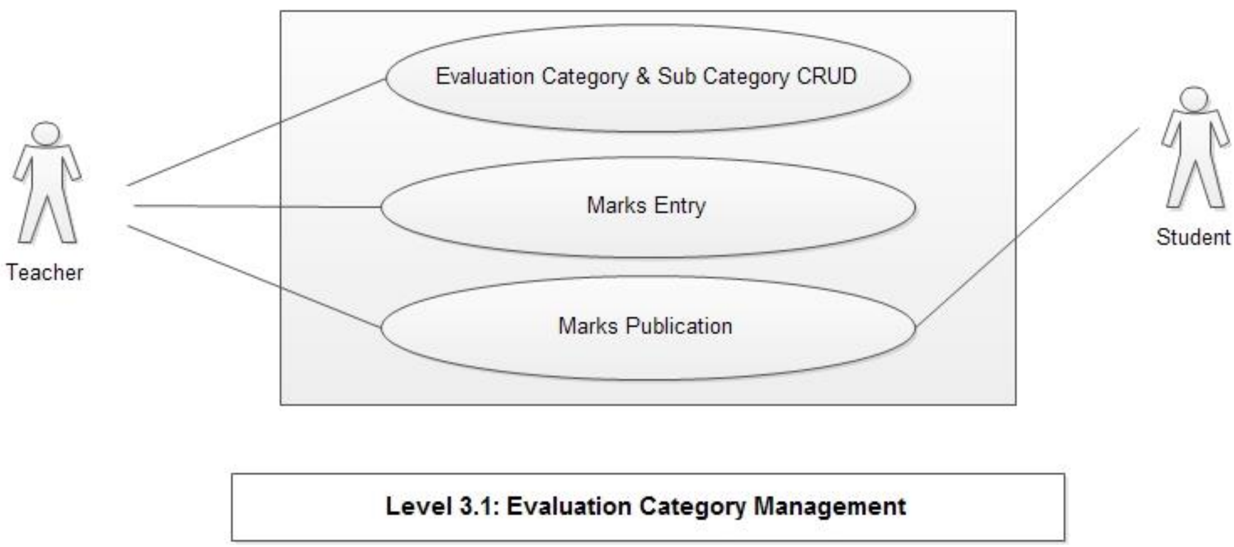

FIGURE 4.2.5: EVALUATION CATEGORY MANAGEMENT (LEVEL 3.1)

Usecase Name: Evaluation Category Management

Usecase ID: 3.1

Primary Actor: Teacher

Secondary Actor: Student

Scenario for level 3.1: Teacher can create categories of marks such as final, mid-term, presentation, project, continuous evaluation and lab. Teacher can also create several sub-categories under each category. He will enter these types of marks in the dashboard. After teachers press “finish” button, students can access their final result from the dashboard after logging in the app.

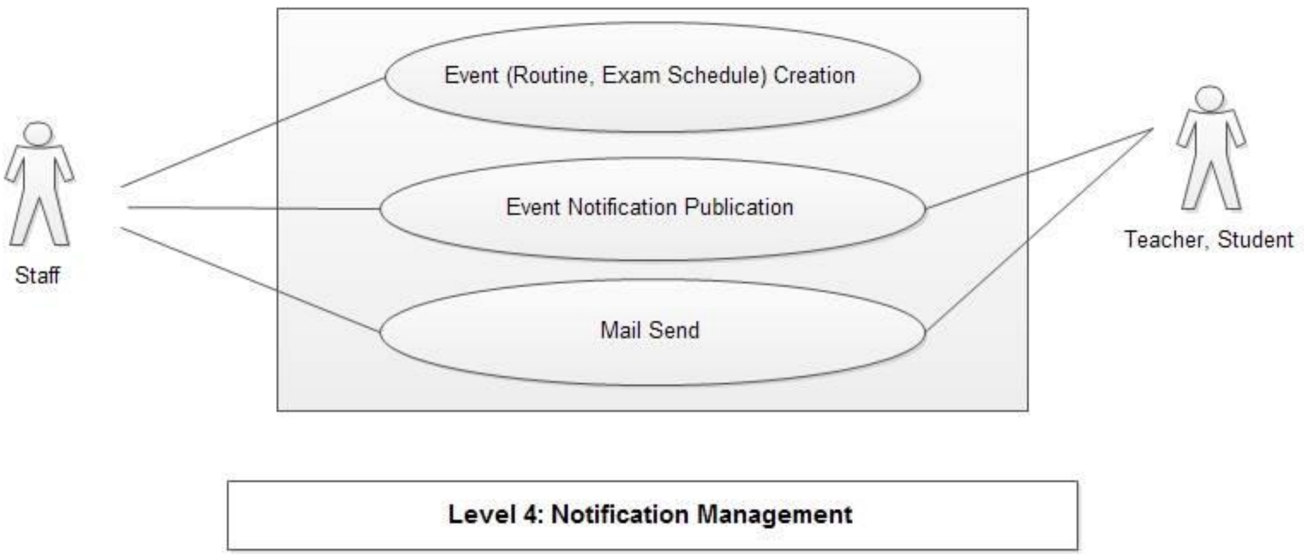

FIGURE 4.2.6: NOTIFICATION MANAGEMENT (LEVEL 4)

Usecase Name: Notification Management

Usecase ID: 4

Primary Actor: Staff, Teacher

Secondary Actor: Student

Scenario for level 4: Staffs can provide notices by uploading files regarding holidays, vacations, exam schedules, routines and others. Event notification about these notices will be sent to intended receivers. Teachers can also give a notice about the next course exam in the dashboard and students of that course will receive immediate notification about the exam.If new notification loads in the notification bar an email will be sent to his/her Gmail account to notify him/her about new notifications.

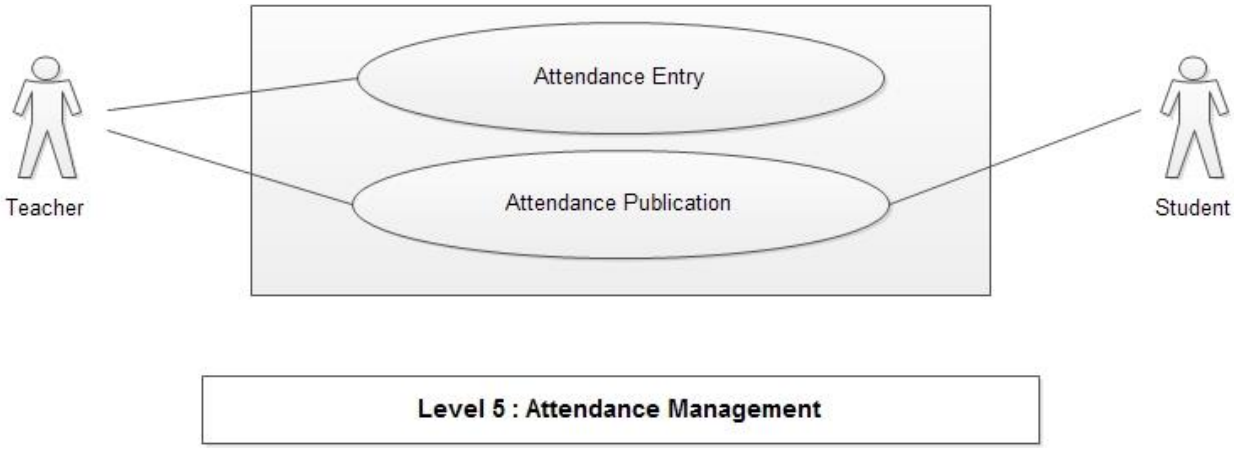

FIGURE 4.2.7: ATTENDANCE MANAGEMENT (LEVEL 5)

Usecase Name: Attendance Management

Usecase ID: 5

Primary Actor: Teacher

Secondary Actor: Student

Scenario for level 5:Teacher can manage attendance of the students from attendance management module of the app. This attendance sheet will only be enabled for students after teacher publishes the attendance.

4.3 Activity Diagram

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. The activity diagrams of our system are given below:

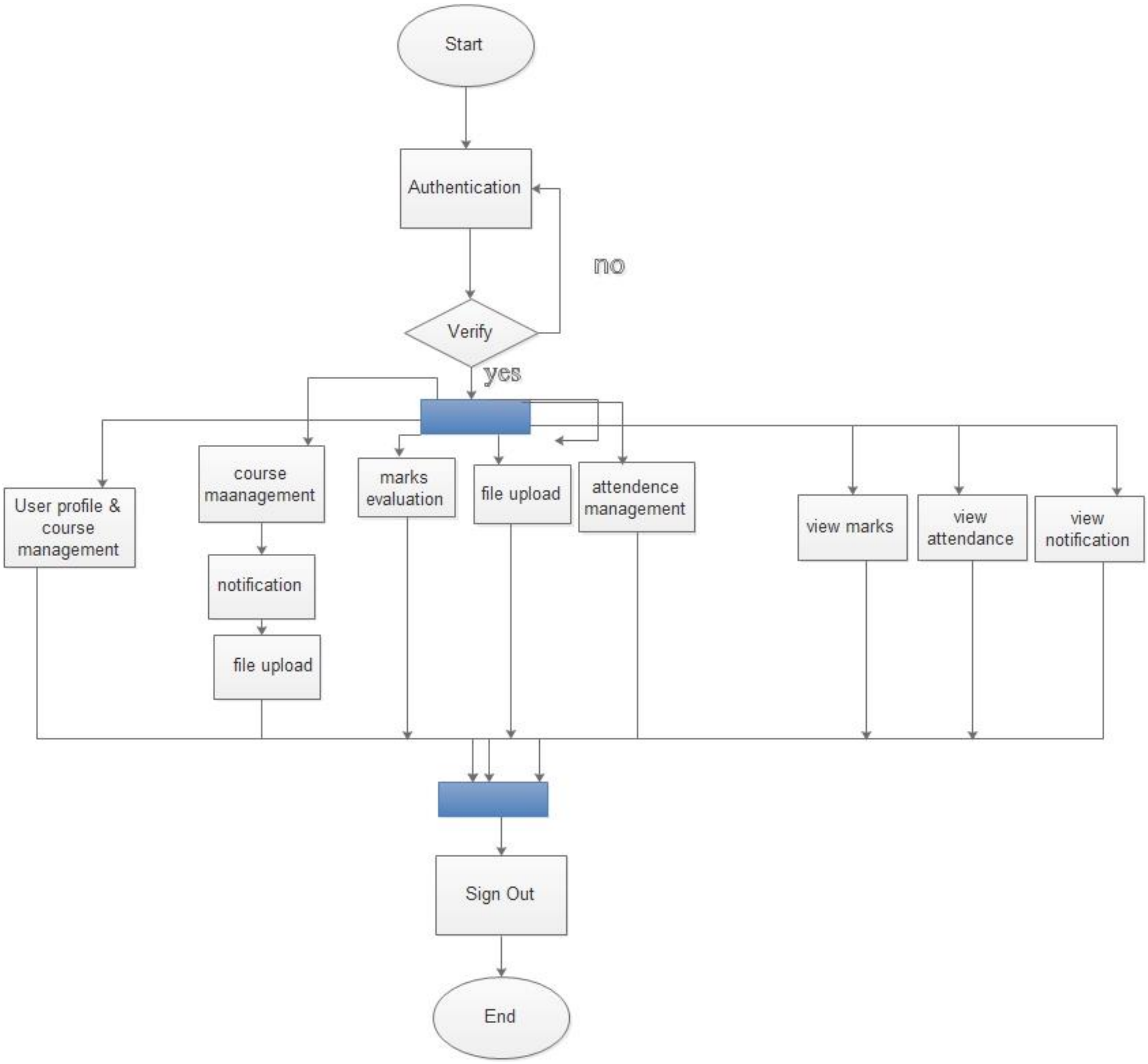

FIGURE 4.3.1: STUDENT INFORMATION MANAGEMENT SYSTEM (LEVEL 0)

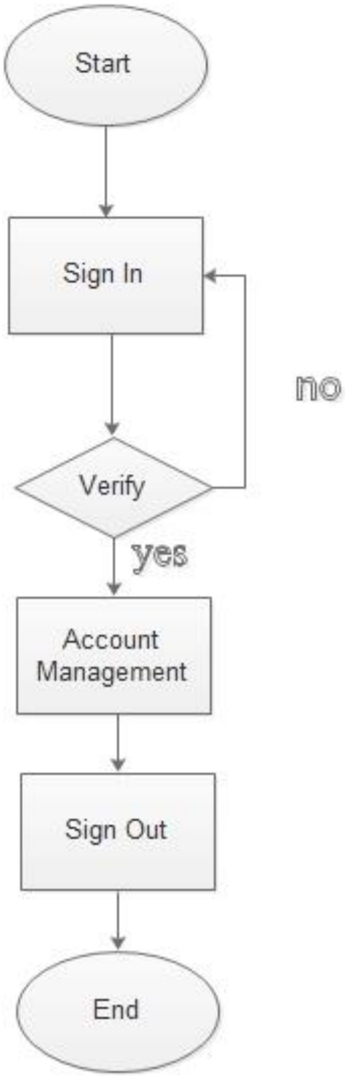

FIGURE 4.3.2: AUTHENTICATION SYSTEM (LEVEL 1)

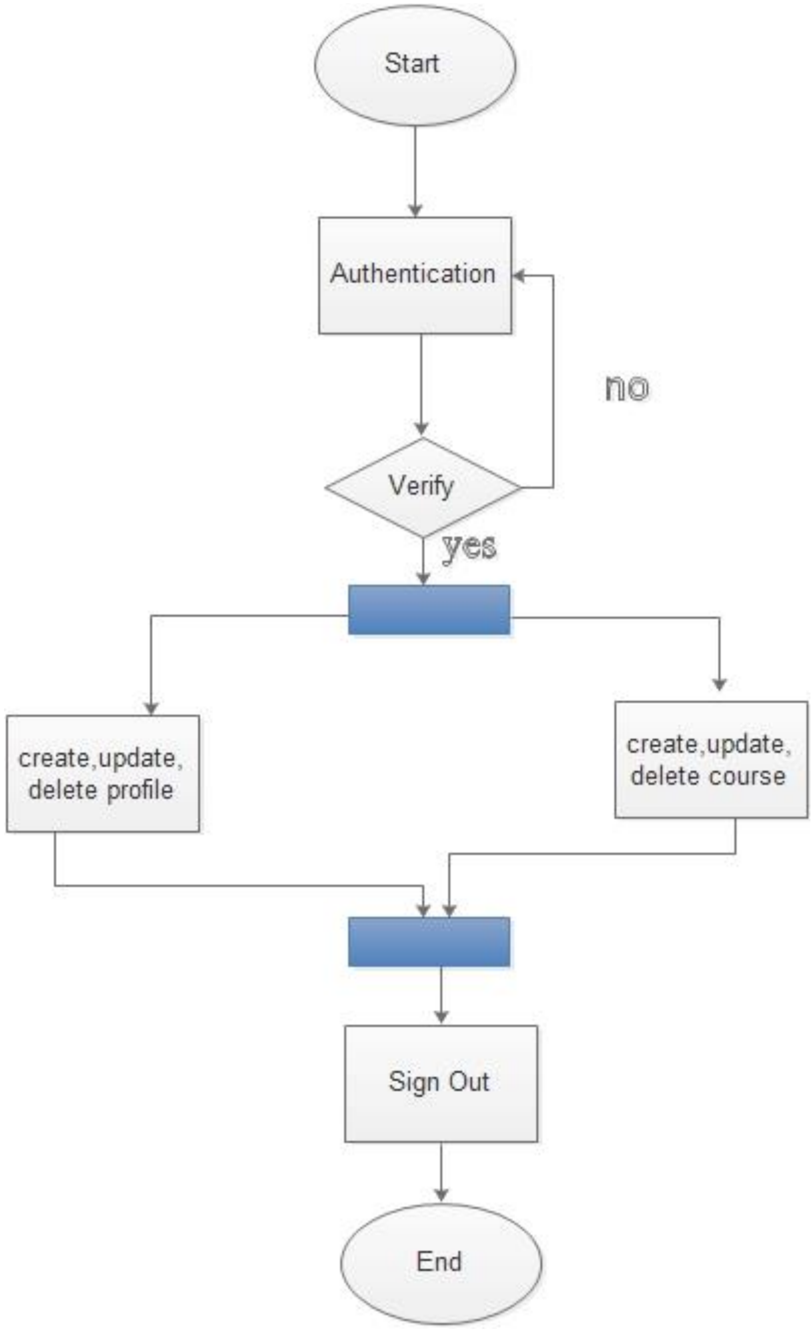

FIGURE 4.3.3: USER PROFILE AND COURSE MANAGEMENT (LEVEL 2)

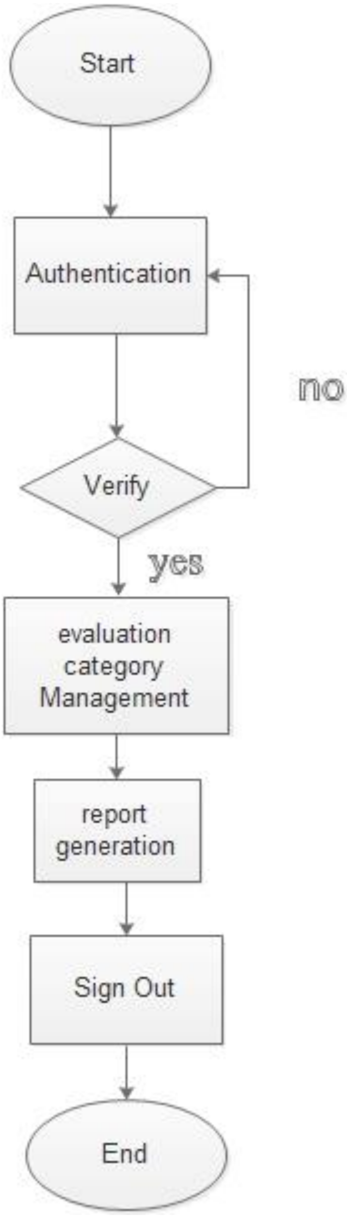

FIGURE 4.3.4: MARKS EVALUATION (LEVEL 3)

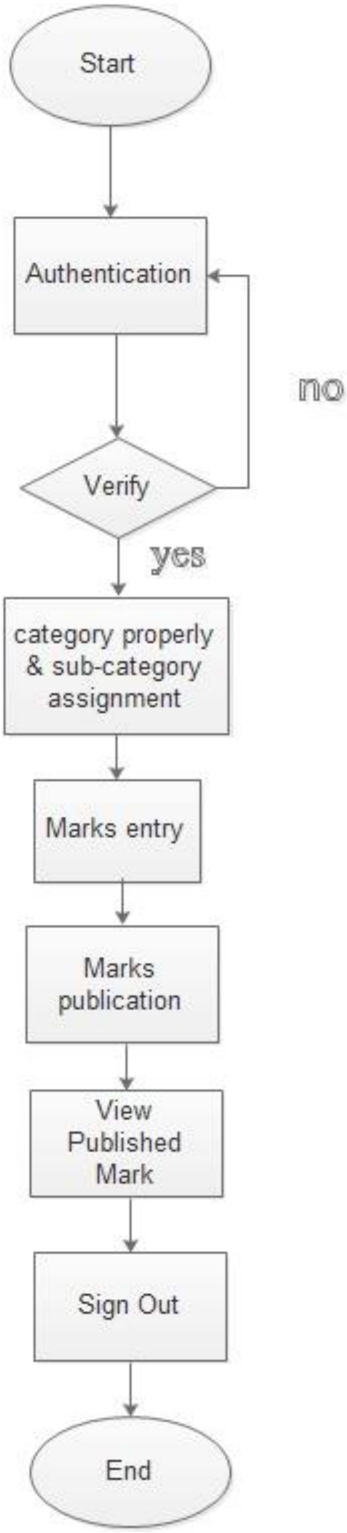

FIGURE 4.3.5: EVALUATION CATEGORY MANAGEMENT (LEVEL 3.1)

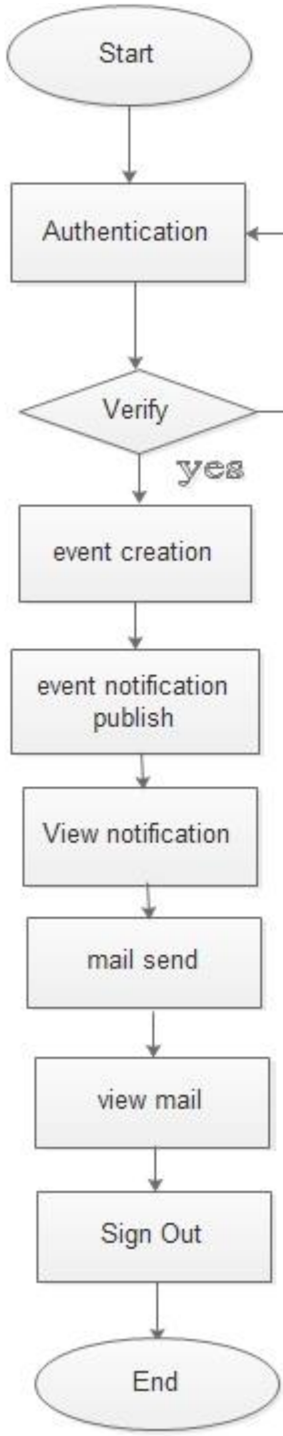

FIGURE 4.3.6: NOTIFICATION MANAGEMENT (LEVEL 4)

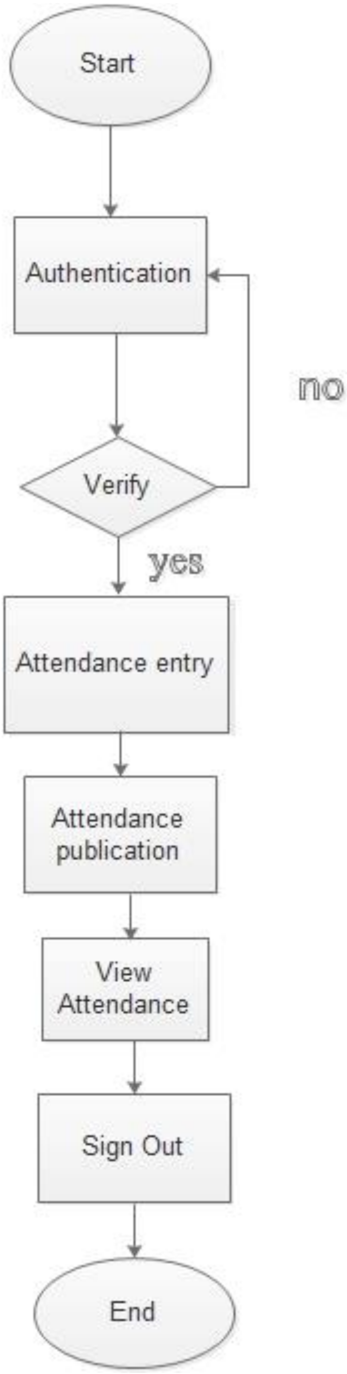

FIGURE 4.3.7: ATTENDANCE MANAGEMENT (LEVEL 5)

4.4 Swimlane Diagram

The swimlane diagrams of our system are given below:

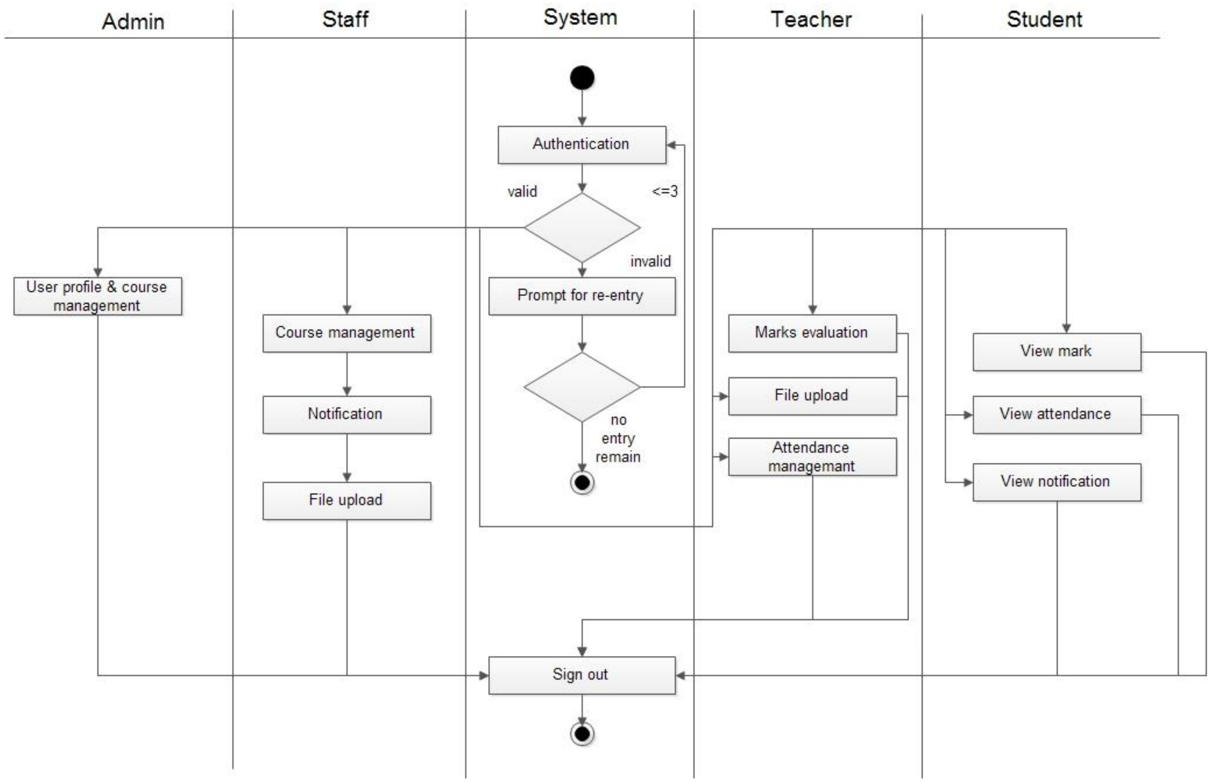

FIGURE 4.4.1: STUDENT INFORMATION MANAGEMENT SYSTEM (LEVEL 0)

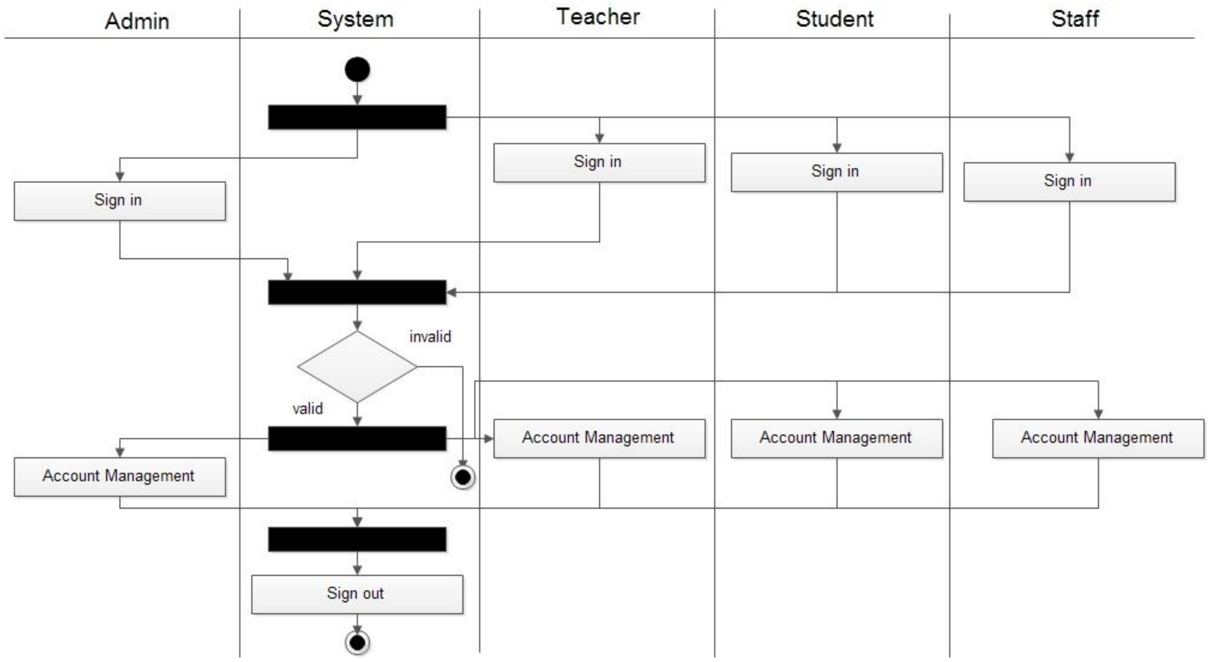

FIGURE 4.4.2: AUTHENTICATION SYSTEM (LEVEL 1)

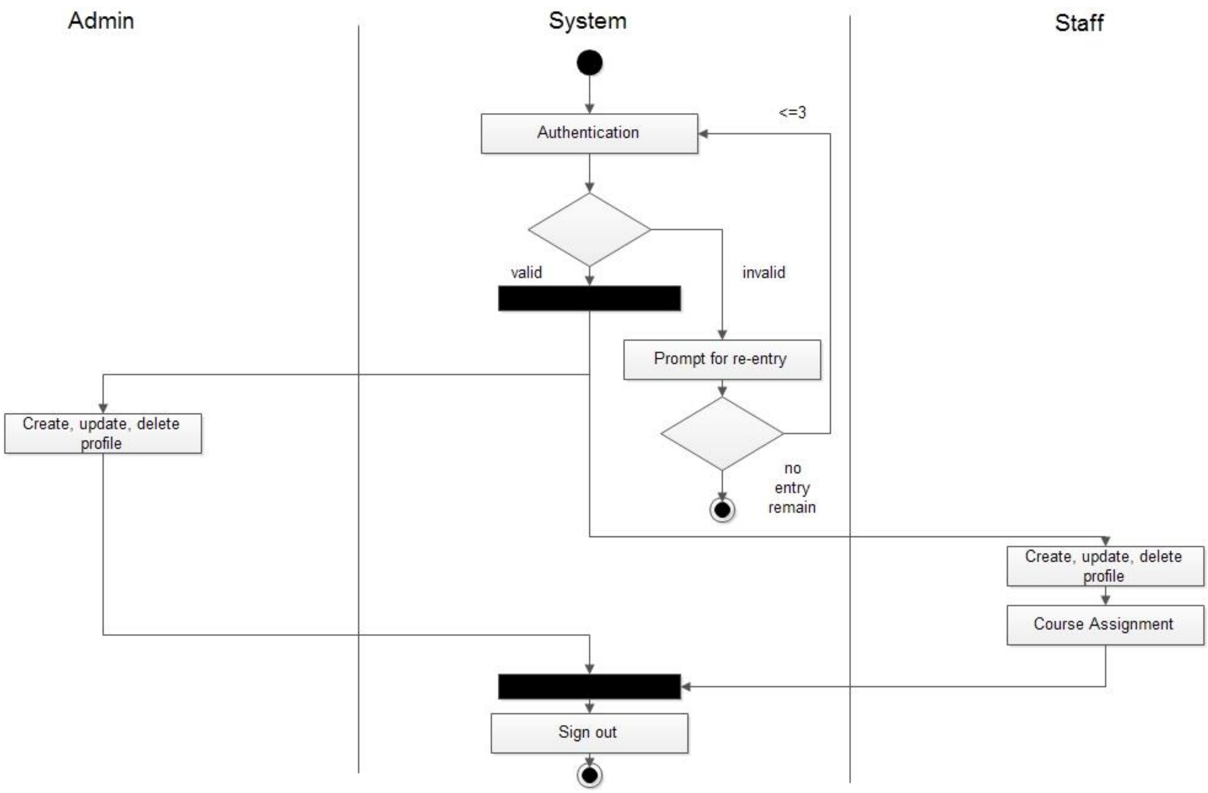

FIGURE 4.4.3: USER PROFILE AND COURSE MANAGEMENT (LEVEL 2)

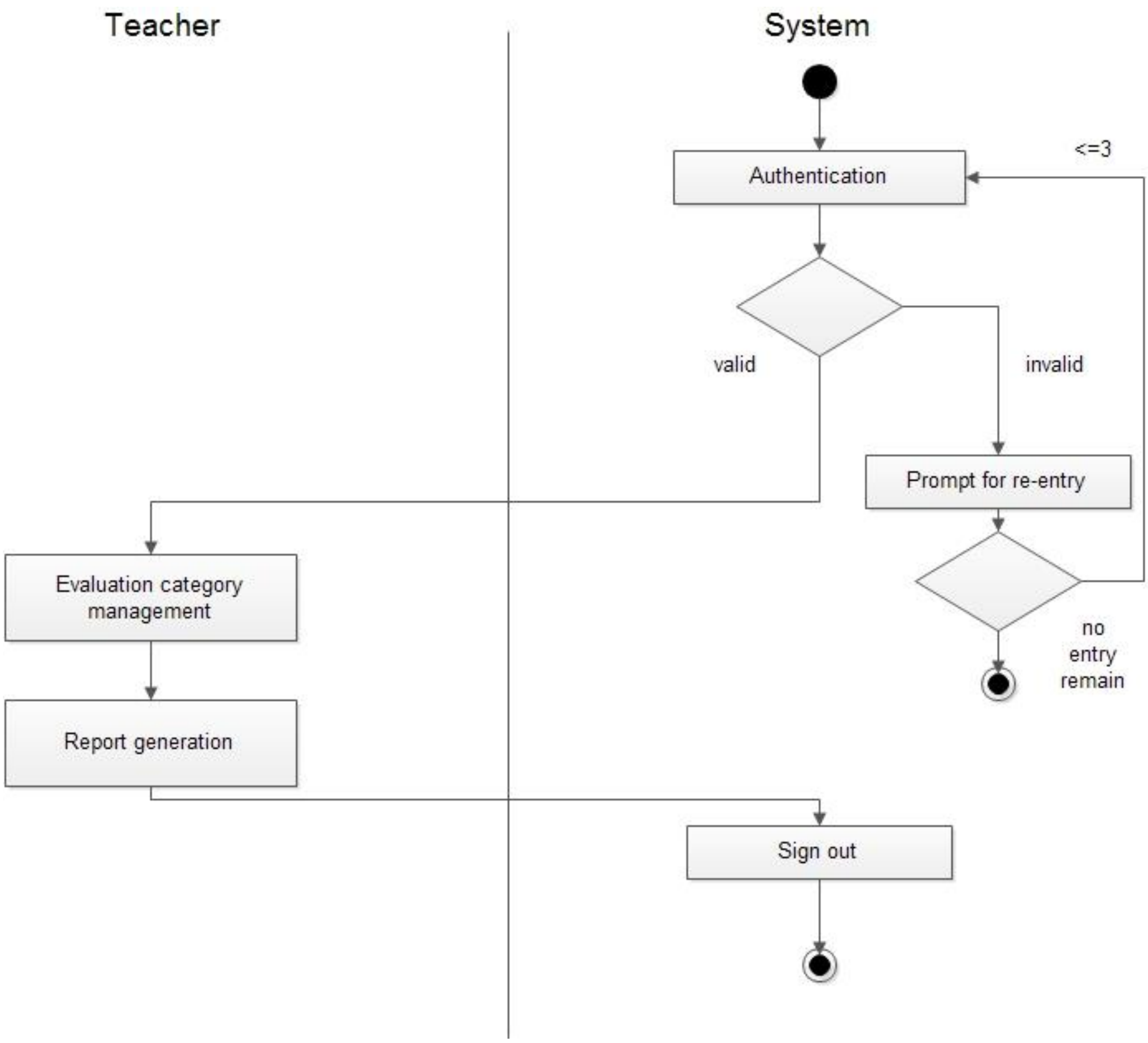

FIGURE 4.4.4: MARKS EVALUATION (LEVEL 3)

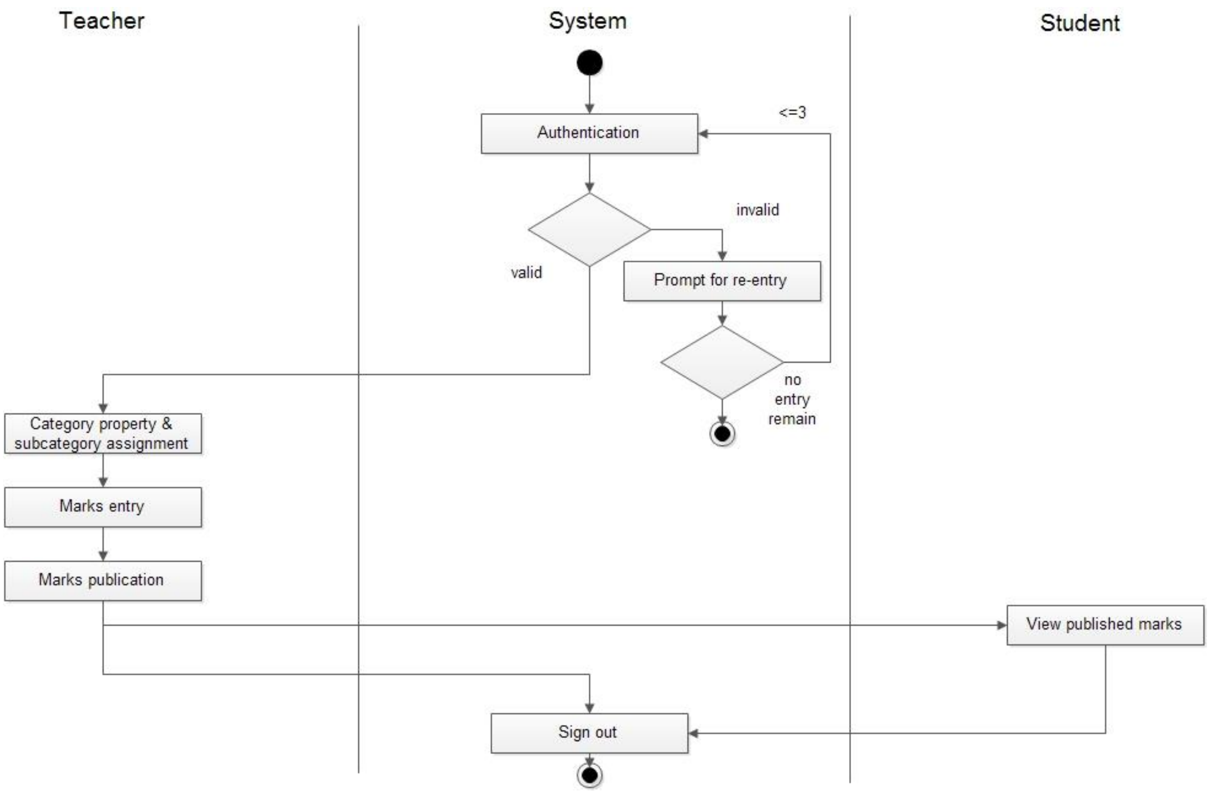

FIGURE 4.4.5: EVALUATION CATEGORY MANAGEMENT (LEVEL 3.1)

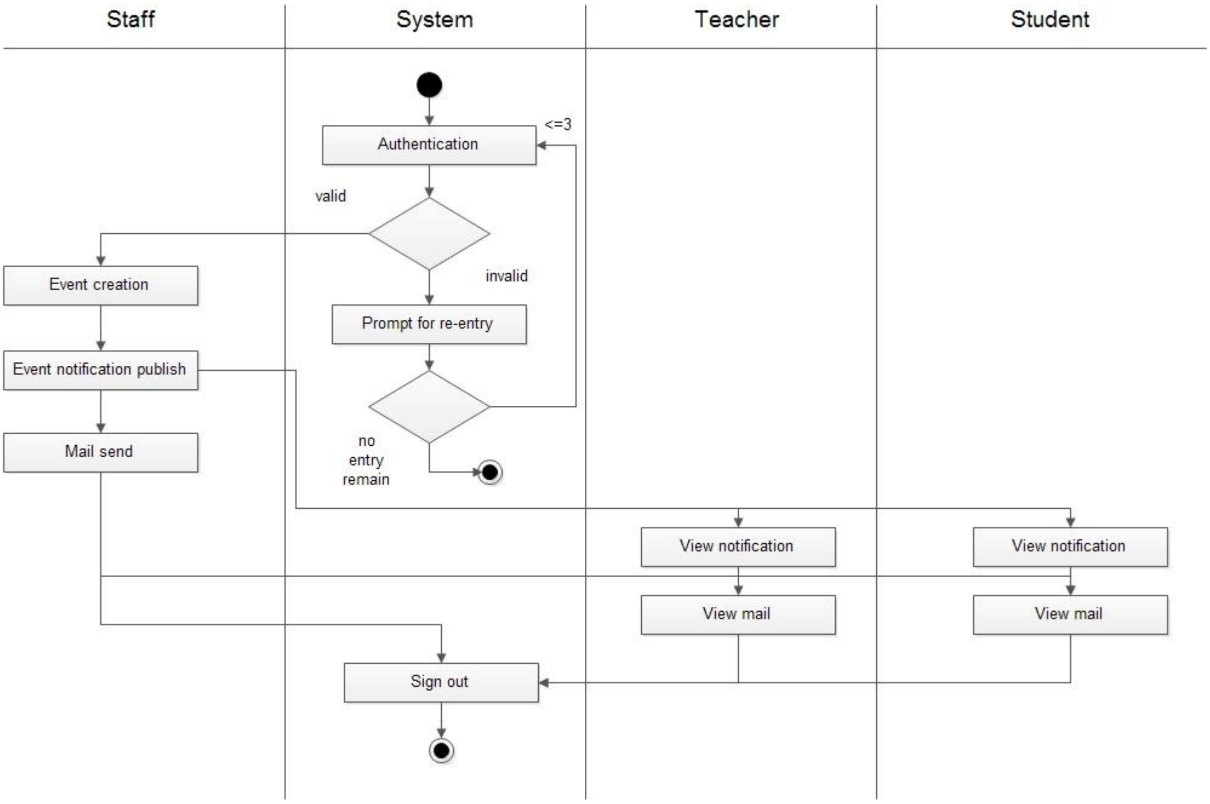

FIGURE 4.4.6: NOTIFICATION MANAGEMENT (LEVEL 4)

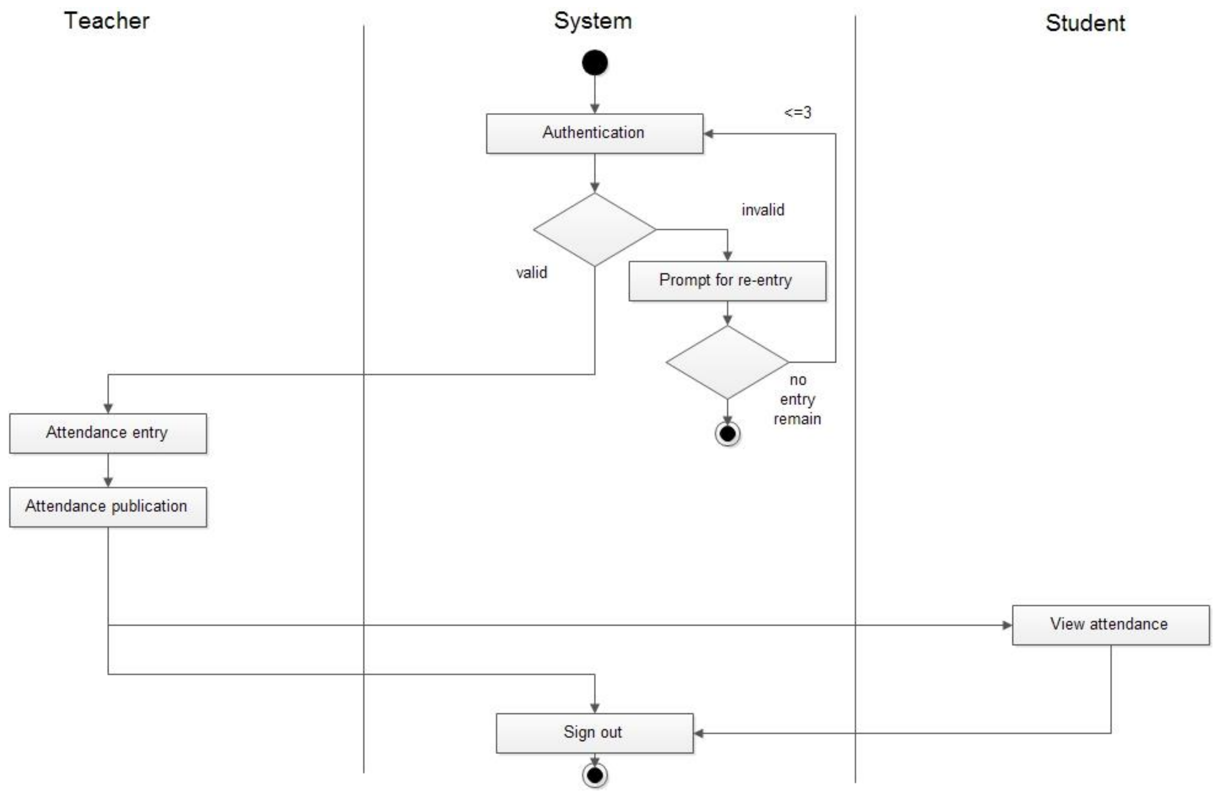

FIGURE 4.4.7: ATTENDANCE MANAGEMENT (LEVEL 5)

4.5 CONCLUSION

We have done this part to develop the scenario. In next chapter the data based model is appeared.

CHAPTER 5

DATA MODEL OF STUDENT INFORMATION MANAGEMENT SYSTEM

5.1 DATA MODELING CONCEPT

Data modeling is a common activity in the software development process of information systems, which typically use database management systems to store information. The output of this activity is the data model, which describes the static information structure in terms of data entities and their relationships. This structure is often represented graphically in entity-relationship diagrams (ERD).

The data model of our project contains important architectural information and serves the following practical purposes:

- Provide a conceptual design of objects in the system's domain and their relationships
- Provide a blueprint for creating the database structure
- Guide implementation of code units that access the database
- Guide performance enhancement in data access operations

A set of following activities is done to design the data model

- Data identification and define attributes for each data
- Create data relationship diagram
- ER diagram
- Table or schema diagram

5.2 DATA IDENTIFICATION AND DEFINE ATTRIBUTES OF EACH DATA

A data object or data is a representation of information which has different properties or attributes that must be understood by the software. To identify a data object all the nouns of the scenario are needed to be analyzed. So we prepared a noun table to identify the data objects.

NOUN TABLE

Noun	Accepted/Rejected	Comments
Server	×	
Students	✓	Views notifications and evaluation marks
Course	✓	Assigned by staffs
Teachers	✓	Performs marks evaluation and generate grade sheet
User	✓	It is the super user that has the common attributes of all users
Account	×	Users account is not necessary as users has the same attributes
Staff	✓	Assign courses
Dashboard	×	It will be developed at the end
Routine	×	Routine will be uploaded in a file format
Admin	✓	Creates user and manage accounts
Event Notification	✓	Staffs will generate it and others will view it
Marks	✓	Teachers will insert marks of a

		student
Attendance	×	There is no action of this in the system
Report	×	Teachers will generate a report of marks
Category	✓	Each course will have category to evaluate marks like lab, assignment etc.
Subcategory	✓	Each category may have subcategories like lab category may contain lab1, lab2
File	✓	File will be uploaded and downloaded by the users
Information	×	Information will be included by the admin so there is no direct connection of this with the system

TABLE 1: NOUN TABLE FROM SCENARIO

From the noun table we identified the data objects and the attributes:

Data Object

Admin

Attributes

- Id
- Email
- Password
- userName
- Role
- contactNumber

Staff

Attributes

- Id
- Email
- Password
- userName
- Role
- contactNumber

Student

Attributes

- Id
- Email
- Password
- userName
- rollNo
- batchNo
- currentYear
- currentSemeater

- contactNumber

Teacher

Attributes

- Id
- Email
- Password
- userName
- Designation
- contactNumber

Course

Attributes

- id
- courseName
- credit
- courseNumber

Event Notification

Attributes

- id
- type
- description
- subject

Category

Attributes

- id
- name
- description
- date
- weight
- isSelected
- hasSubcategory

Subcategory

Attributes

- category_id
- id
- name
- description
- date
- marksOutof

File

Attributes

- id
- name
- location

- size
- description

MARKS

ATTRIBUTES

- id
- listifMarks

REPORT

ATTRIBUTES

- id
- creatorName
- date
- subject
- body

5.3 ENTITY RELATIONSHIP DIAGRAM AND SCHEMA DIAGRAM

ER Diagram shows the total picture of the interactions of data and it facilitates to construct schema and database deigning of the system.

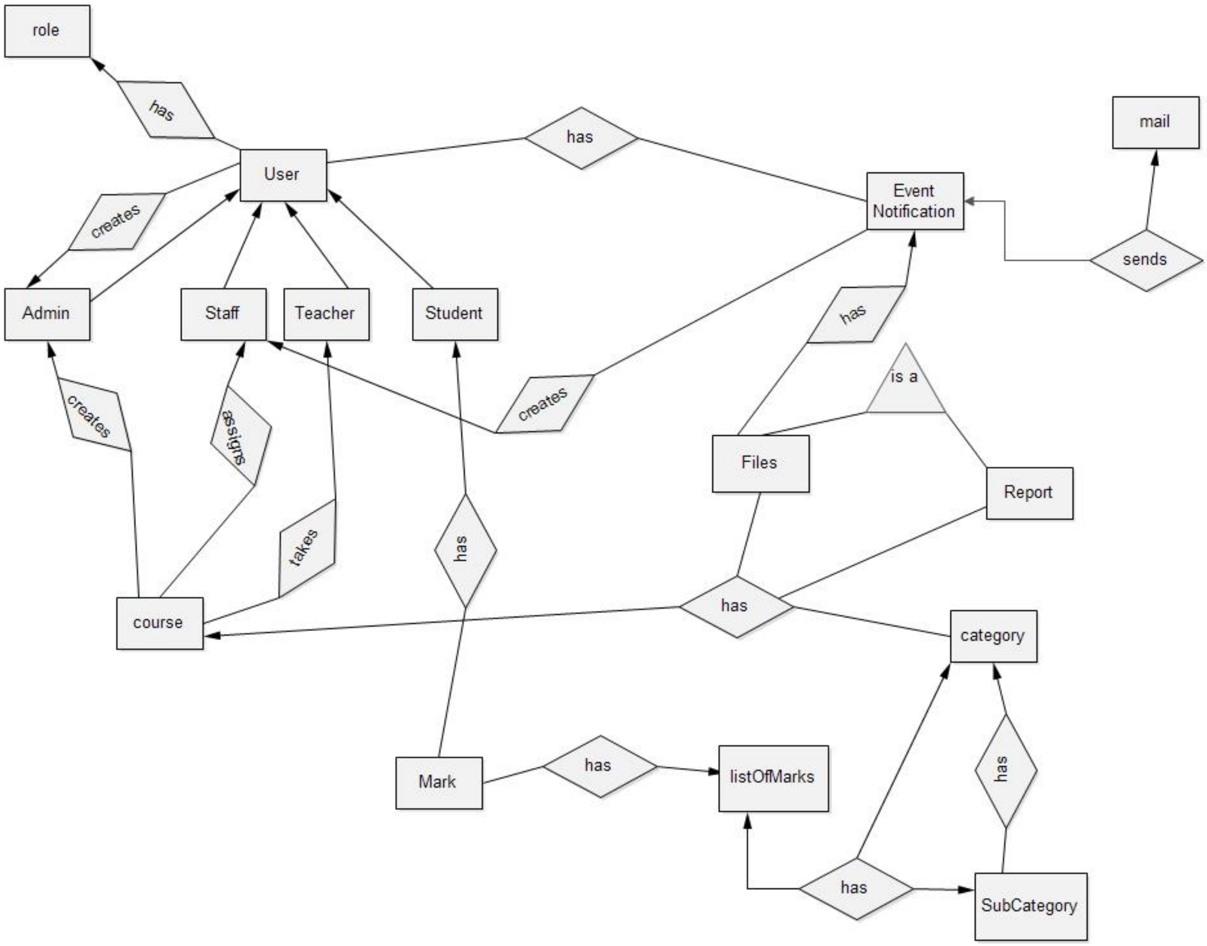

FIGURE 5.3.1: ER DIAGRAM

SCHEMA

The schema diagram is shown in next page

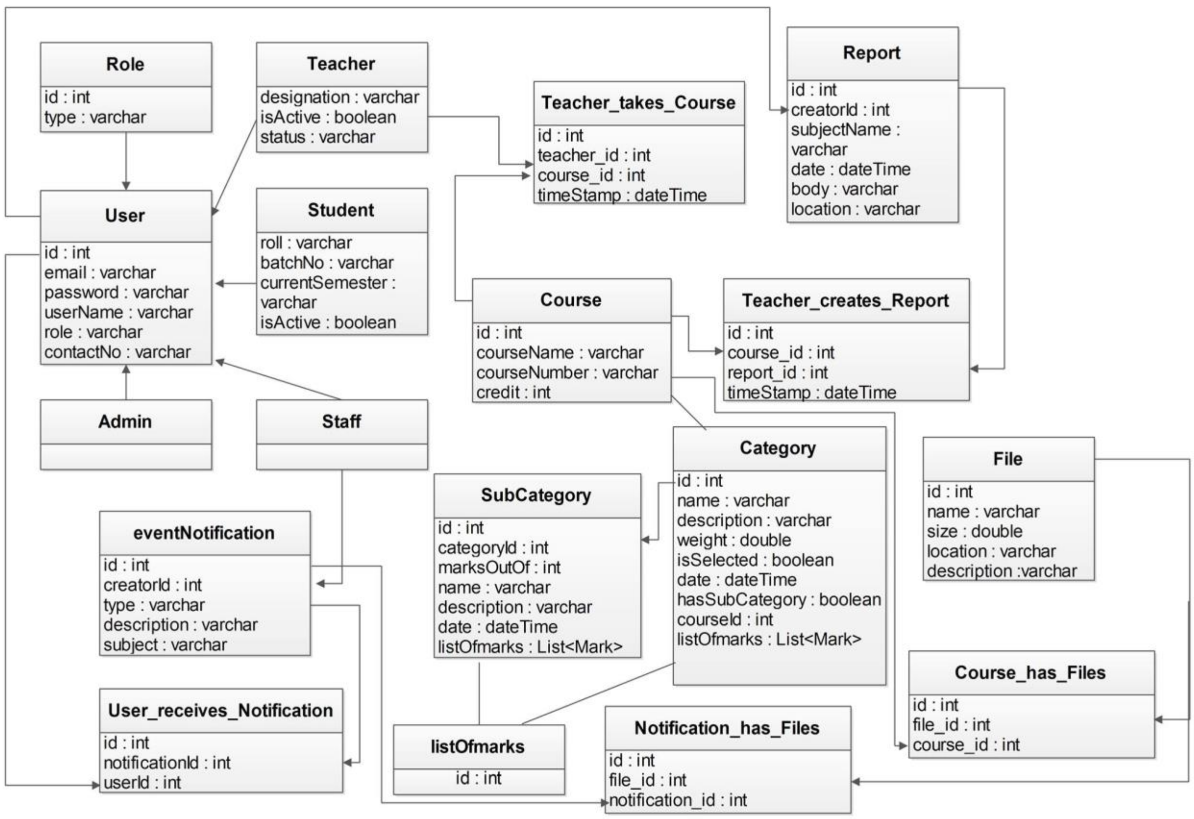

FIGURE 5.3.2: SCHEMA DIAGRAM

5.4 CONCLUSION

In this chapter we have identified all the nouns and their attributes. We have established relationships among data objects. In next chapter we will describe the class model.

CHAPTER 6

This Chapter is intended to describe class based modeling of student information management system.

CLASS BASED MODELING CONCEPT

Class-based modeling represents the objects that the system will manipulate, the operations that will applied to the objects, relationships between the objects and the collaborations that occur between the classes that are defined. The elements of a class-based model include classes and objects, attributes, operations, class-responsibility-collaborator (CRC) models, class diagrams.

6.1 IDENTIFYING ANALYSIS CLASSES

Examining all the nouns from the usage scenario potential classes can be identified. For class identification we have to go through the following steps.

STEP 1

Identifying and categorize all nouns in following ways:

- **External Entities** (e.g. other systems, devices, people) that produce or consume information to be used by a computer-based-system.
- **Things** (e.g. report, displays, letters, signals) that are part of the information domain for the problem.
- **Occurrence or events** (e.g. a property transfer or the completion of a series of robot movements) that occur within the context of system operation.
- **Roles** (e.g. manager, engineer) played by people who interact with the system.
- **Organizational units** (e.g. division, group, and team) those are relevant to an application.
- **Places** (e.g. manufacturing floor or loading dock) that establish of the problem and the overall function of the system.
- **Structures** (e.g. sensors, vehicles or computer) that define a class of objects or related classes of object.

STEP 2

Selection of potential class is performed by considering six selection characteristics.

- **Retained information:** The potential class will be useful during analysis only if information about it must be remembered so that the system can function.
- **Needed services:** The potential class must have a set of identifiable operations that can change the value of its attributes in some way.
- **Multiple attributes:** During requirement analysis, the focus should be on “major” information; a class with a single attribute may, in fact, be useful during design, but is probably better represented as an attribute of another class during the analysis activity.
- **Common attributes:** A set of attributes can be defined for the potential class and these attributes apply to all instances of the class.
- **Common operations:** A set of operations can be defined for the potential class and these operations apply to all instances of the class.
- **Essential requirements:** External entities that appear in the problem space and produce or consume information essential to the operation of any solution for the system will almost always be defined as classes in the requirements model.

6.2 POTENTIAL CLASS TABLE:

Potential class	General Classification	Characteristics
User	Roles	Accepted all applied
Course	External entity, roles	Accepted all applied
Event_Notification	External entity	Accepted all applied
File	Structure	Accepted all applied
Marks	Structure	Accepted all applied
Report	Things	Accepted all applied
Category	Structure	Accepted all applied
SubCategory	Structure	Accepted all applied

TABLE 2: POTENTIAL CLASS IDENTIFICATION

From this table we found the following potential classes:

- User(Admin, Staff, Teacher, Student)
- Course
- Event_Notification
- File
- Marks
- Report
- Category
- SubCategory

6.3 SPECIFYING ATTRIBUTES

Attributes describe a class that has been selected for inclusion in the requirements model. In essence, it is the attributes that define the class—that clarify what is meant by the class in the context of the problem space.

We can represent the attributes of selected classes in the following manner:

Admin = id + email + password + userName + role + contactNumber

Staff = id + email + password + userName + role + contactNumber

Student = roll + batchNo + currentYear + currentSemester + id + email + userName + password + contactNumber

Teacher = id + email + password + designation + userName + contactNumber

Course= id + courseName + credit + courseNumber

Event_Notification= id + type + description + subject

File = id + name + size + location + description

Marks = id + List<Mark>

Report = id + creatorName + date + body + subject

Category = id + description + name + weight + isSelected + date + hasSubcategory

Subcategory= categoryId + id + marksOutOf + description + name + date

6.4 Defining Operations

Operations define the behavior of an object. The operations or methods are given below:

Admin = create_account(), create_role(), delete_account(), store_information()

Staff = assign_course(), provide_notice(), provide_routine()

Student = view_attendace(), view_marks(), download(), receive_notification(), ask() ,get_email()

Teacher = provide_notice(), set() ,update(), delete(), see_course(), set_evaluation(), receive_notification(), manage_attendance(), get_email()

Course = get(), set(), update(), delete()

EventNotification = get(), set(), send()

File = get(), set(), upload(), download()

Marks = get(), set(), update(), delete()

Report = create(), send(), view()

Category = get(), set(), update(), delete(), selected(), weighted()

Subcategory = get(), set(), update(), delete(), selected(), weighted()

6.5 Class Diagram for Student Management Information System

Admin
Id **Email** **Password** **User_name** **Role** **contactNumber**
create_account(); **create_role();** **delete_account();** **store_information();**

Staff

Id
Email
Password
User_name
Role
contactNumber

assign_course()
provide_notice()
provide_routine()

Student

Roll
Batch_no
Current_year
Current_Semester
Id
Email
user_name
password
contactNumber

view_attendace();
view_marks();
download();
receive_notification();
ask();
get_email();

Teacher

Id
Email
Password
Designation
user_name contactNumber

provide_notice()
set()
update()
delete()
see_course(), set_evaluation()
receive_notification()
manage_attendance()
get_email()

Course

Id
Course_Name
Credit
Course_number

get()
set()
update()
delete()

EventNotification

Id
Type
Description
Subject

get()
set()
send()

File

Id
Name
Size
Location
Description

get()
set()
upload() download()

Marks

Id
List_of_marks

get()
set()
update()
delete()

Report

Id
Creator_Name
Date
Body
Subject

create()
send()
view()

Category

Category
Id
Description
Name
Weight
isSelected
Date
hasSubcategory
get()
set()
update()
delete()
selected()
weighted()

SubCategory

SubCategory
Id
Category_id
marks_out_of Description
Name
Date
get()
set()
update()
delete()
selected()
weighted()

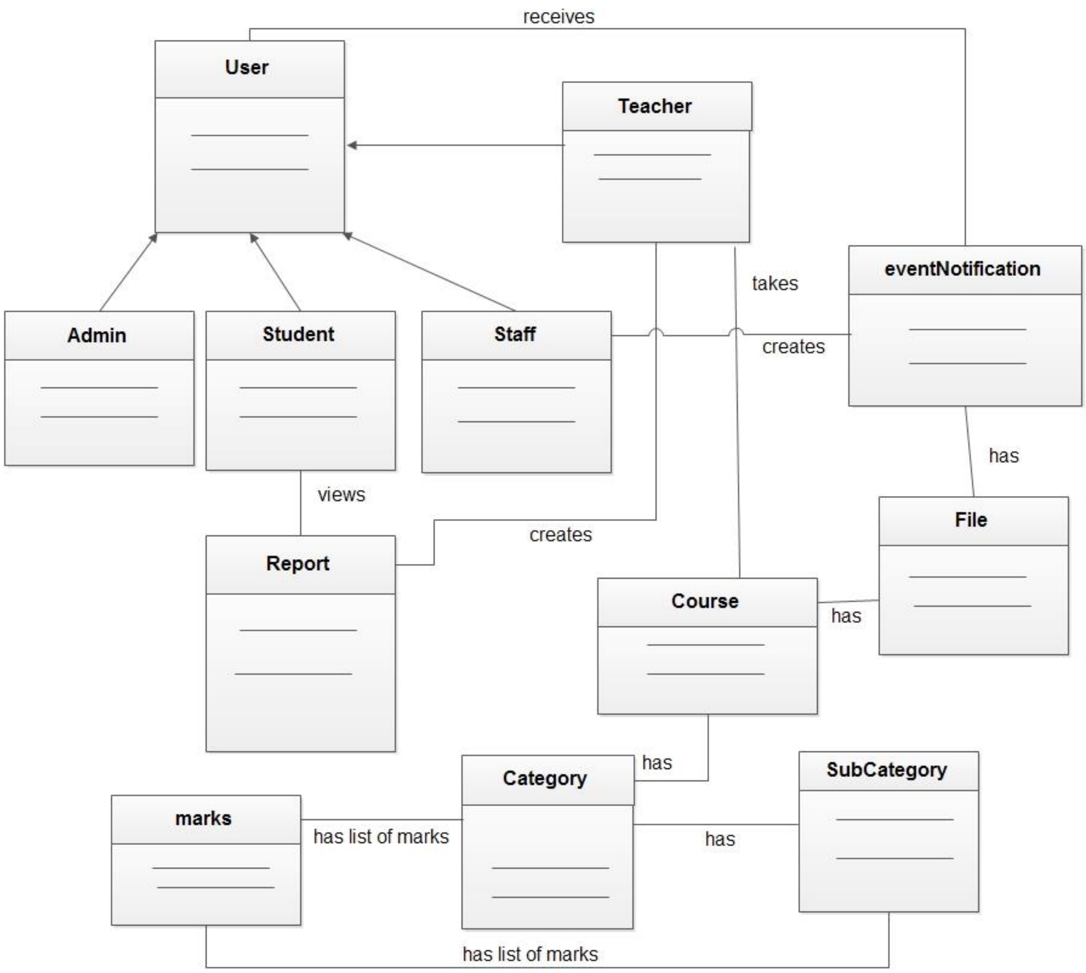

FIGURE 6.5.1: CLASS DIAGRAM

6.6 Class Responsibility Collaborator (CRC)

Class name: Admin

Responsibility	Collaborator
1.Create Account	Staff, Teacher, Student
2.Create Role	Staff, Teacher, Student
3. Delete Account	Staff, Teacher, Student
4. Store information	Course

Class name: Staff

Responsibility	Collaborator
1.Assign course	Teacher, Student
2.Provide notice	Teacher, Student
3.Provide routine	Teacher, Student
4.Send email	

Class name: Teacher

Responsibility	Collaborator

1.Provide notice	Student
2.Set evaluation	Student
3.Manage attendance	Student

Class name: Student

Responsibility	Collaborator
1.View courses	Course
2.Ask questions	Teacher

Class name: File

Responsibility	Collaborator
1.Upload	Teacher
2.Downlaod	Student

Class name: Marks

Responsibility	Collaborator
1.Give	Teacher
2.Receive	Student

Class name: Event notification

Responsibility	Collaborator
1.Send	Staff
2.Receive	Teacher, Student

Class name: Category

Responsibility	Collaborator
1.Selected	Teacher
2.Weighted	Teacher
3.Take	Student

Class name: Subcategory

Responsibility	Collaborator
1.Selected	Teacher
2.Weighted	Teacher
3.Take	Student

6.7 Conclusion

We have identified all potential classes, drawn class diagram and developed CRC. Next chapter is about dataflow model.

CHAPTER 7

BEHAVIORAL MODEL OF STUDENT INFORMATION MANAGEMENT SYSTEM

7.1. INTRODUCTION

Behavioral model describes the control structure of a dynamic system. This can be things like:

- Sequence of operations
- Object states and
- Object interactions

The behavioral model indicates how our system will response to external events. Here, we first identified all the events from the user scenario, then we drew state diagram for analysis class and we drew sequence diagram.

7.2. IDENTIFYING EVENTS WITH THE USE CASE

We identified the following events from the scenario:

- Admin creates user profile
- Admin assigns roles (teacher, student and staff)
- Admin deletes/deactivates profile
- Teacher uploads files
- Teacher manages and publishes attendance
- Teacher views notification
- Teacher selects courses from course list
- Teacher selects/creates/edits categories/sub-categories
- Teacher entries and publishes marks
- Student views notification, marks and attendance
- Student downloads files
- Staff creates and assigns courses
- Staff creates and sends notification
- Staff sends email

The table for the events identification is:

Event	Initiator	Collaborator
Creating user profile	Admin	Staff, Student, Teacher
Assigning roles (teacher, student and staff)	Admin	Staff, Student, Teacher
Deleting/deactivating profile	Admin	Staff, Student, Teacher
Uploading files	Teacher	
Managing attendance	Teacher	Student
Publishing attendance	Teacher	Student
Viewing notification	Teacher, Student	Staff
Selecting/creating/editing categories/sub-categories	Teacher	Staff
Entering marks	Teacher	Student
Publishing marks	Teacher	Student
Downloading files	Student	Teacher
Viewing marks	Student	Teacher
Viewing attendance	Student	Teacher
Creating and assigning courses	Staff	Teacher
Creates and sending notification	Staff	Teacher, Student
Sending email	Staff	Teacher, Student

TABLE 3: EVENTS IDENTIFICATION

7.3. State Diagram for Analysis Classes

One component of a behavioral model is a state diagram that represents active states for each class and the events that cause changes between these active states. We consider two different characterization of state:

a) The state of each class as the system perform its function

b) The state of the system as observed from the outside as the system performs its function.

The state of the class takes on both active and passive state. State diagram for the analysis class of our system is given below:

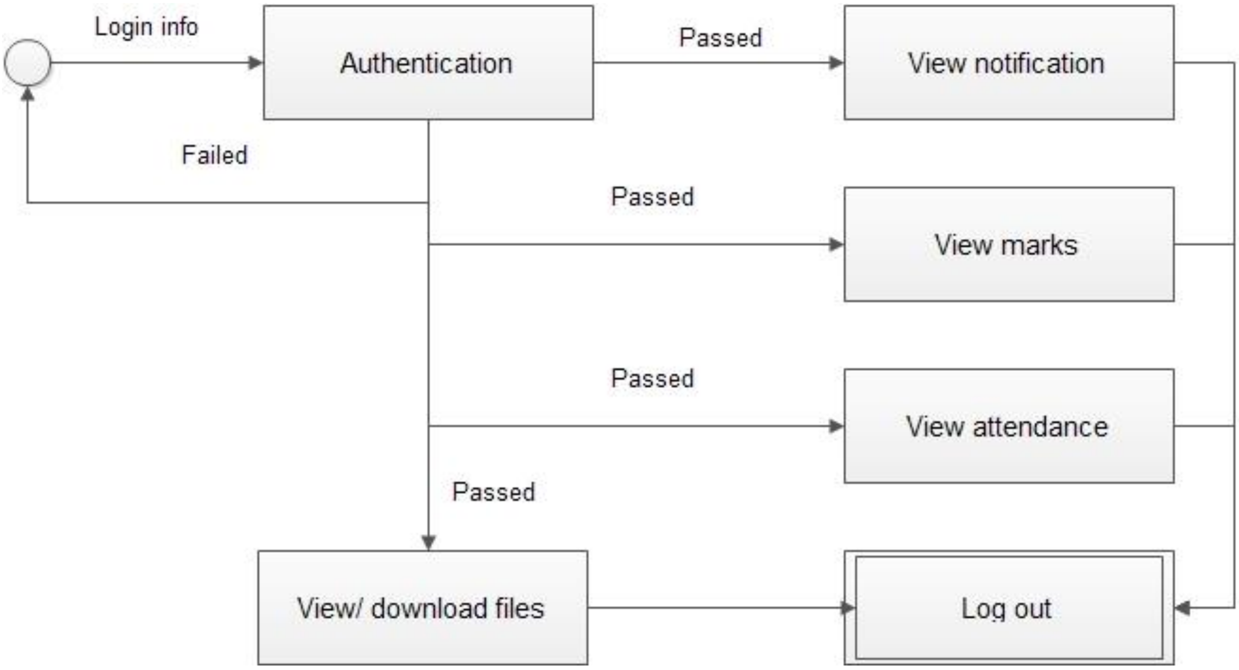

FIGURE 7.3.1: STUDENT

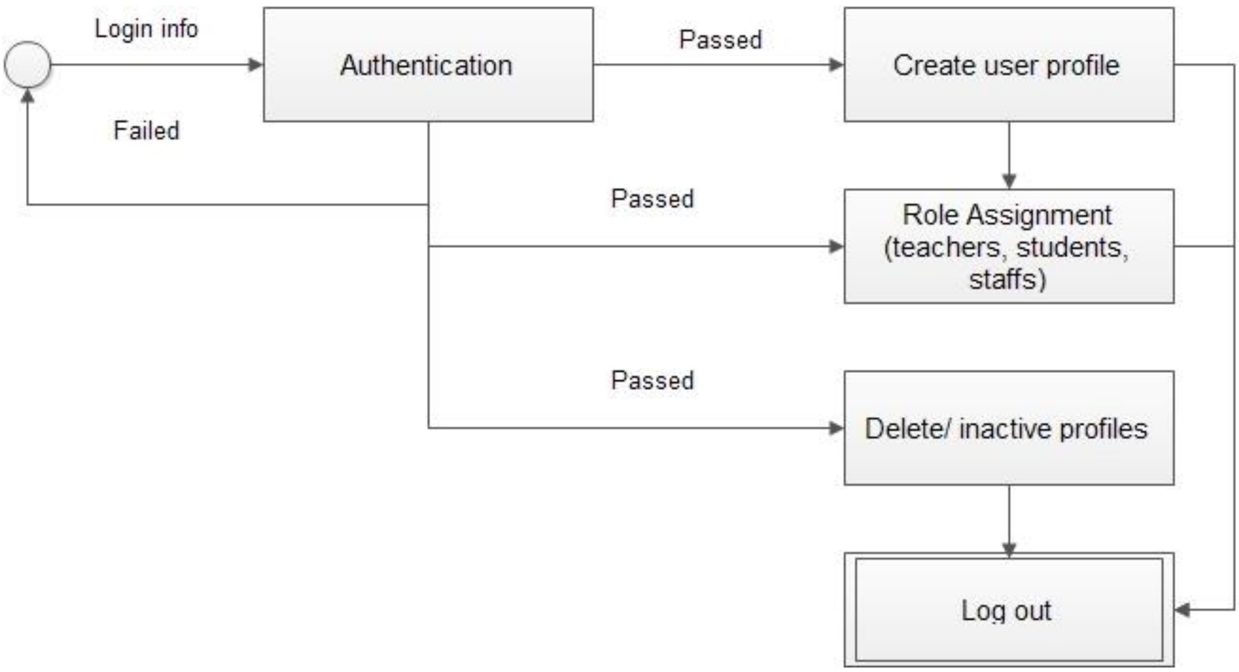

FIGURE 7.3.2: ADMIN

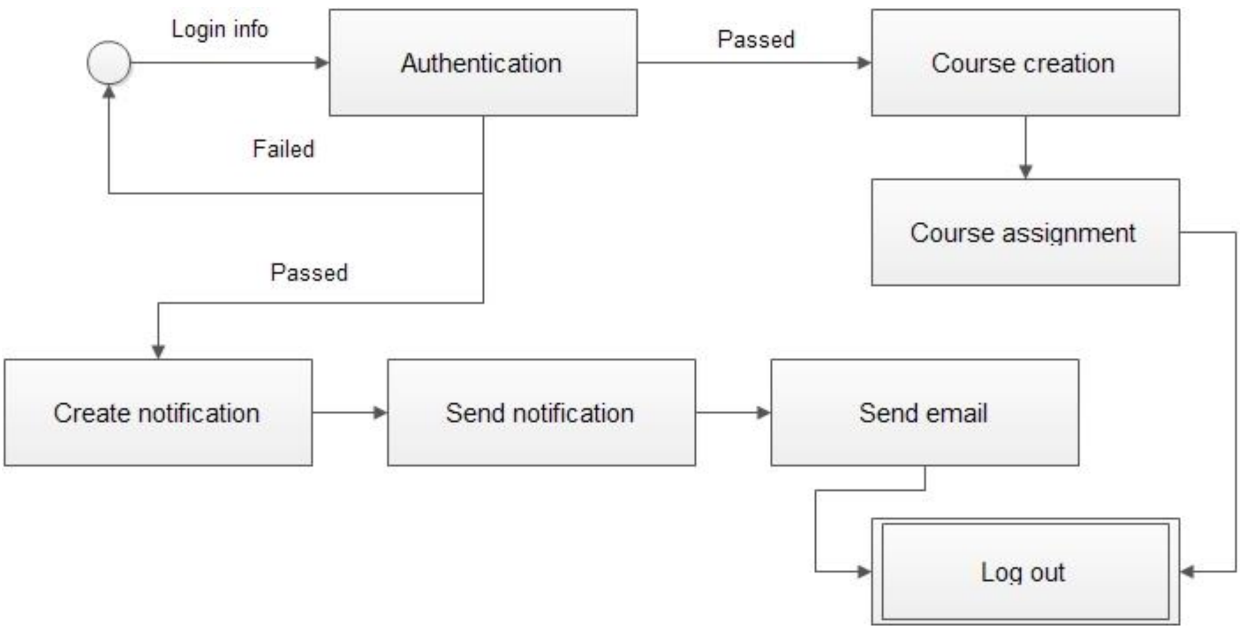

FIGURE 7.3.3: STAFF

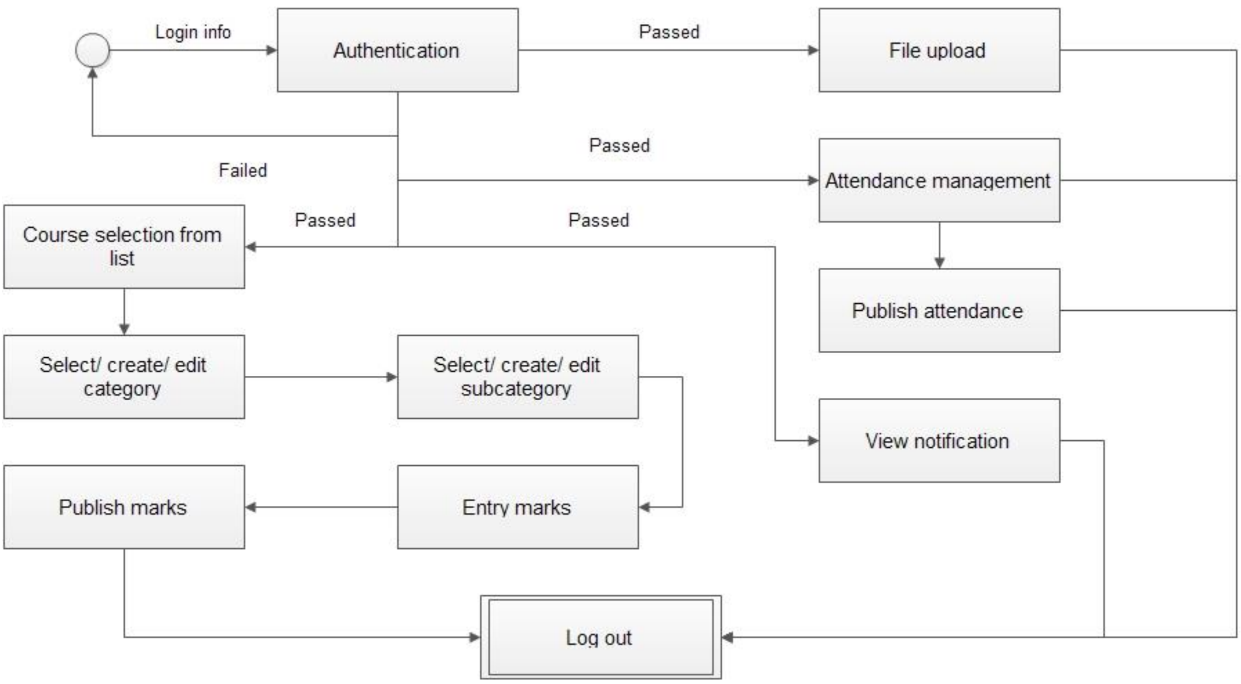

FIGURE 7.3.4: TEACHER

7.4. SEQUENCE DIAGRAM

Sequence diagram indicates how events cause transitions from object to object. Here is only one sequential diagram for our project. The sequence diagram of Student Information Management System is:

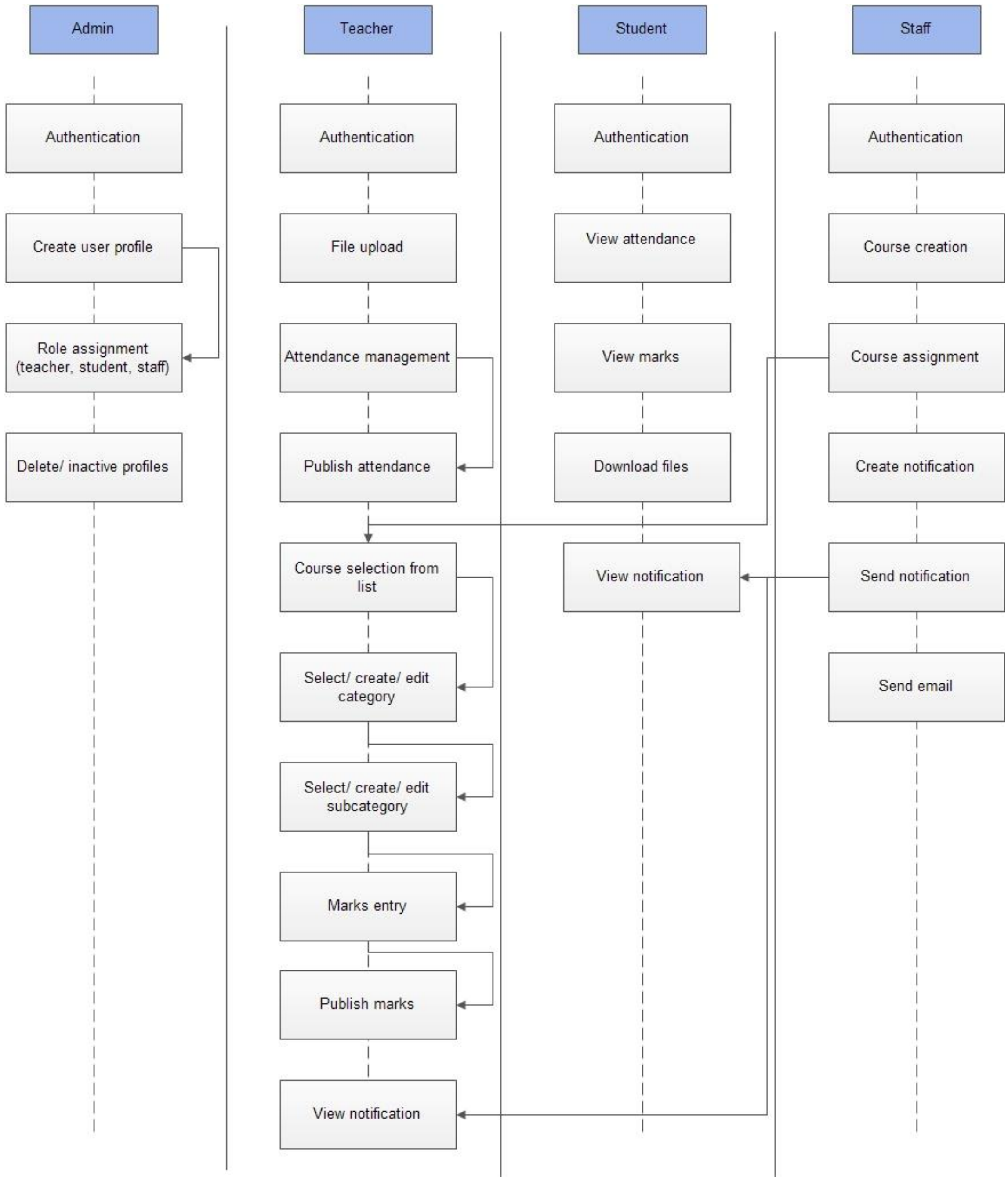

FIGURE 7.4.1: SEQUENCE DIAGRAM

7.5 CONCLUSION

In this chapter we have identified all events. Based on these events we have drawn state diagram and sequence diagram.

CHAPTER 8

MOCK UI

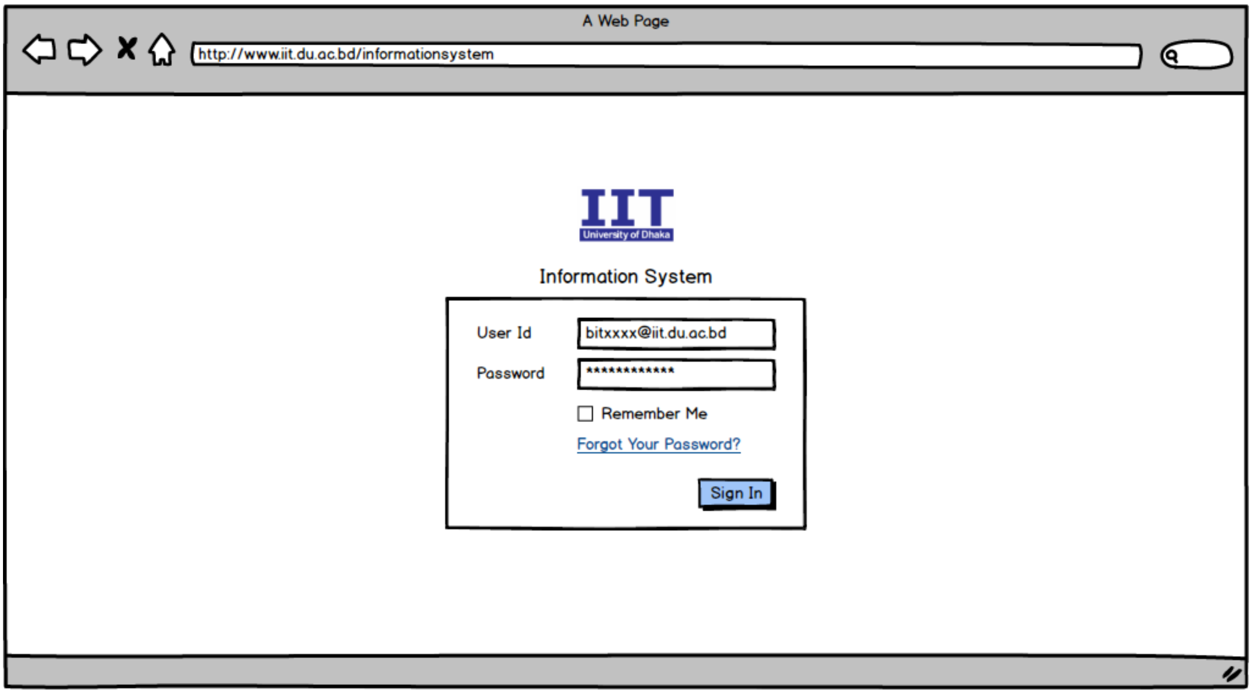

FIGURE 8.8: SIGN IN

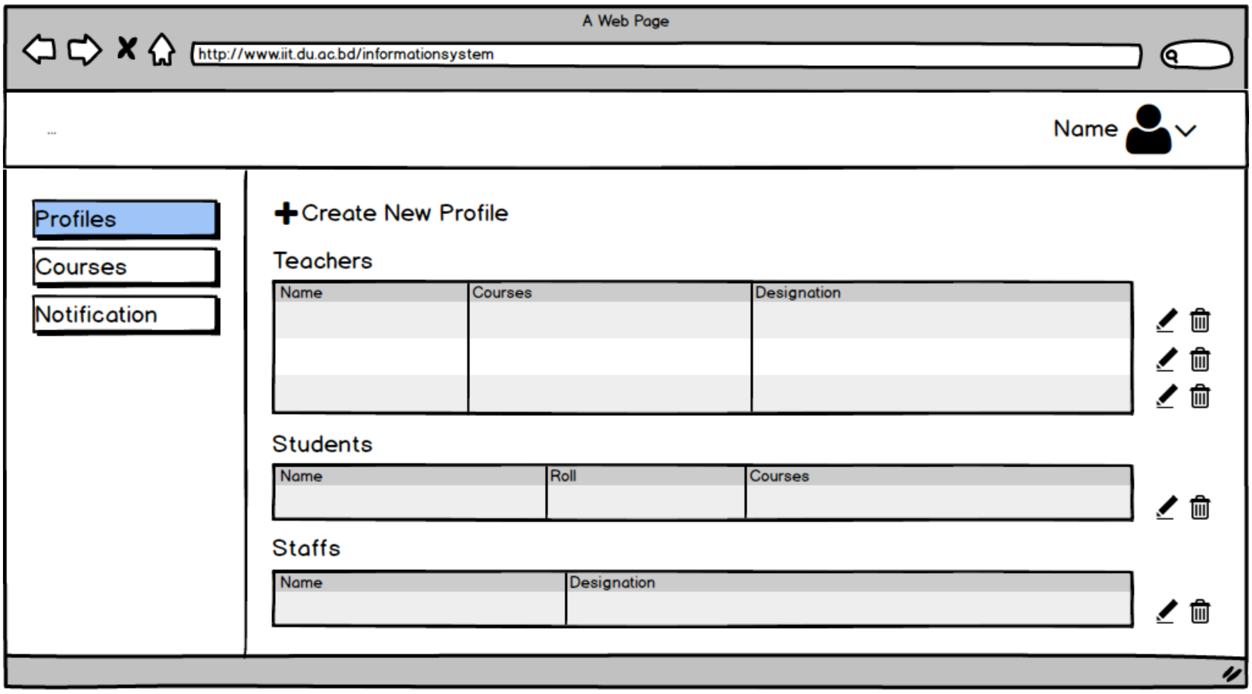

FIGURE 8.9: ADMIN PROFILE 1ST PAGE

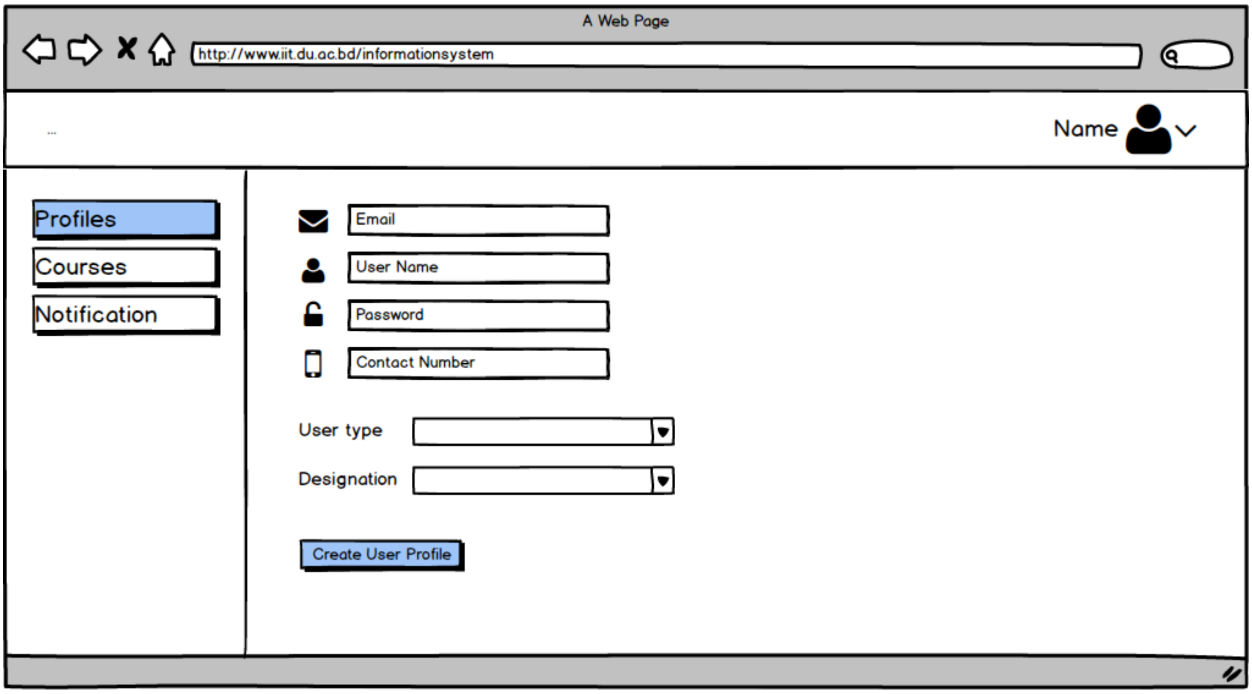

FIGURE 8.10: ADMIN PROFILE 2ND PAGE

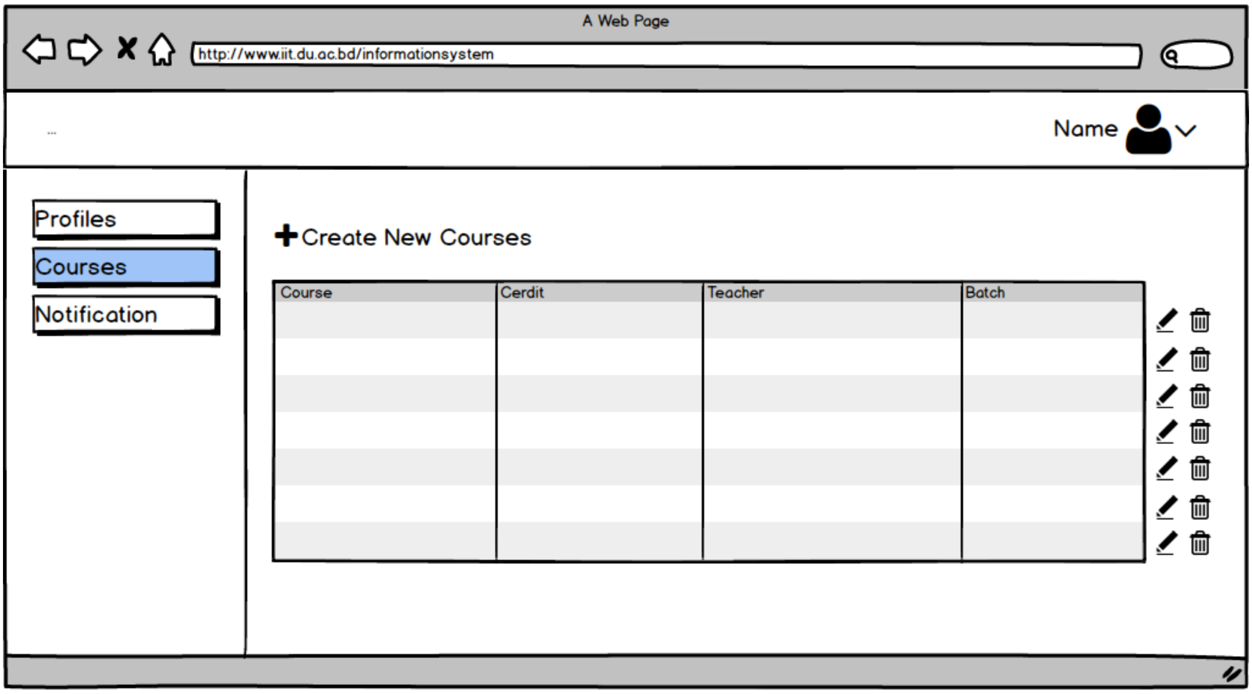

FIGURE 8.11: ADMIN COURSE 1ST PAGE

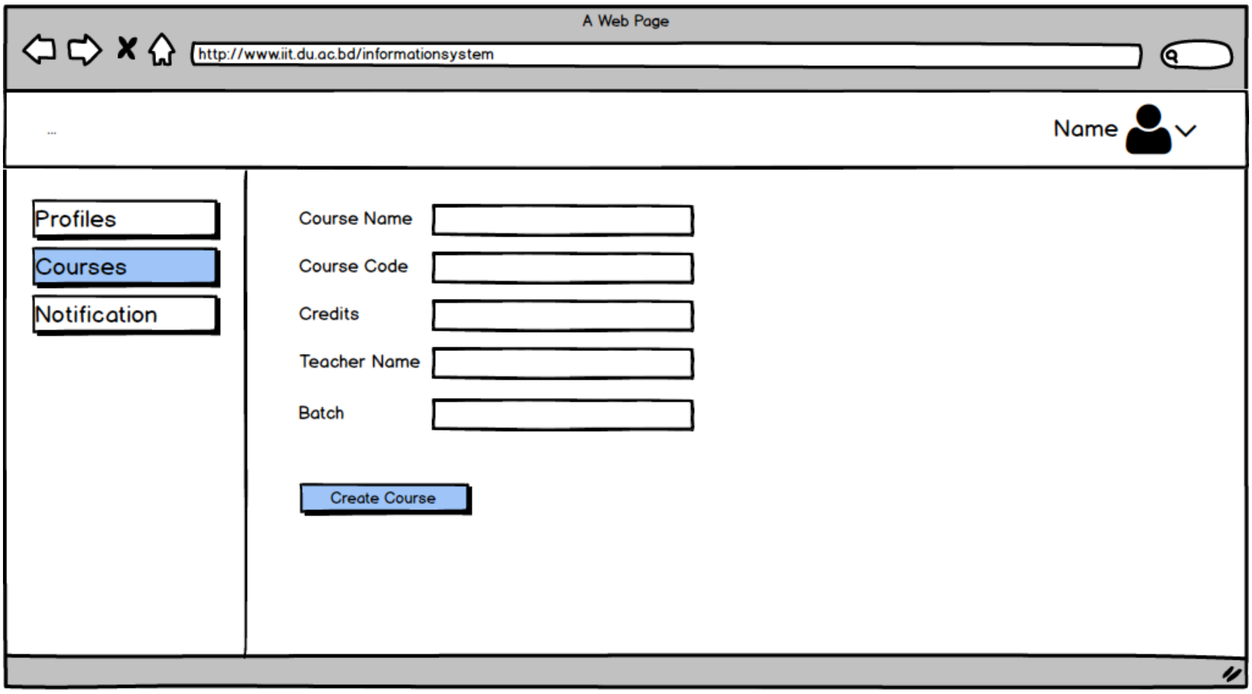

FIGURE 8.12: ADMIN COURSE 2ND PAGE

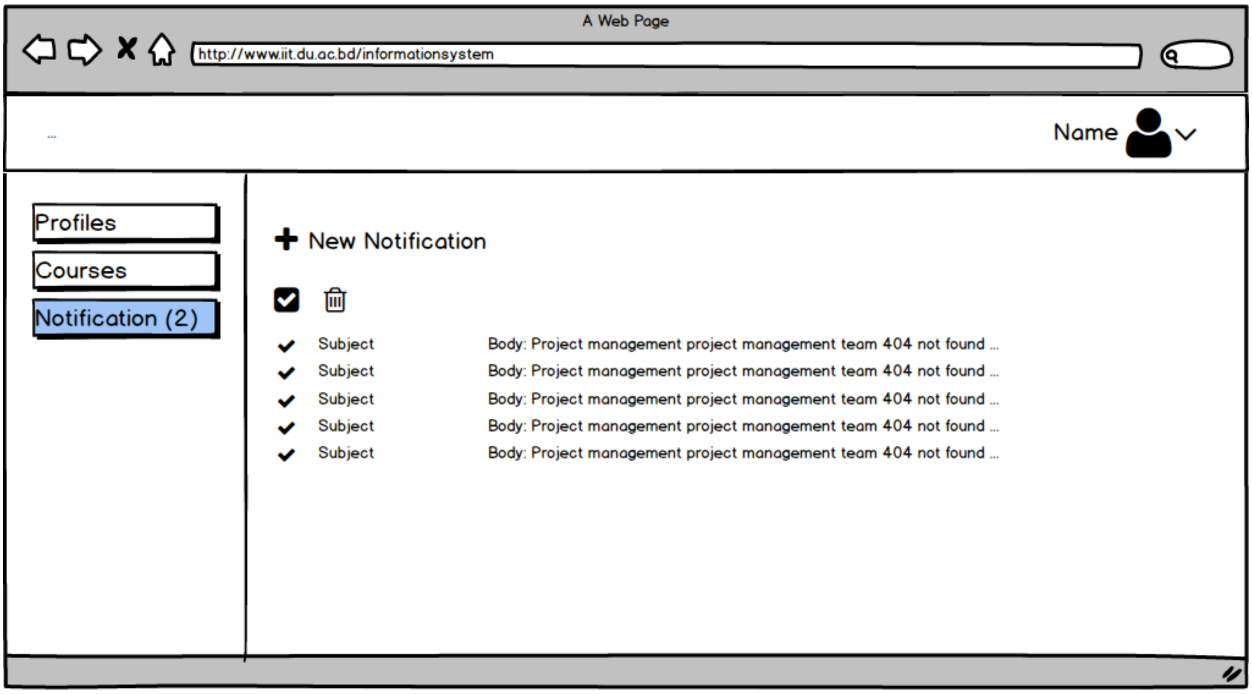

FIGURE 8.13: ADMIN NOTIFICATION 1ST PAGE

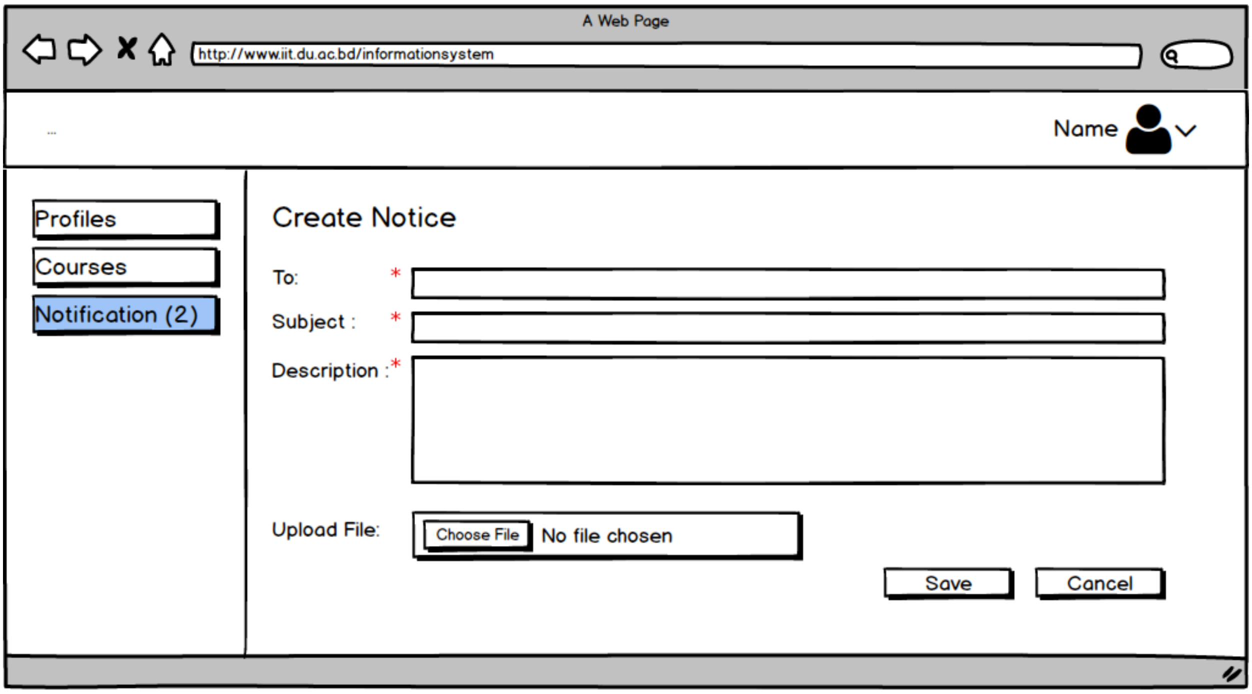

FIGURE 8.14: ADMIN NOTIFICATION 2ND PAGE

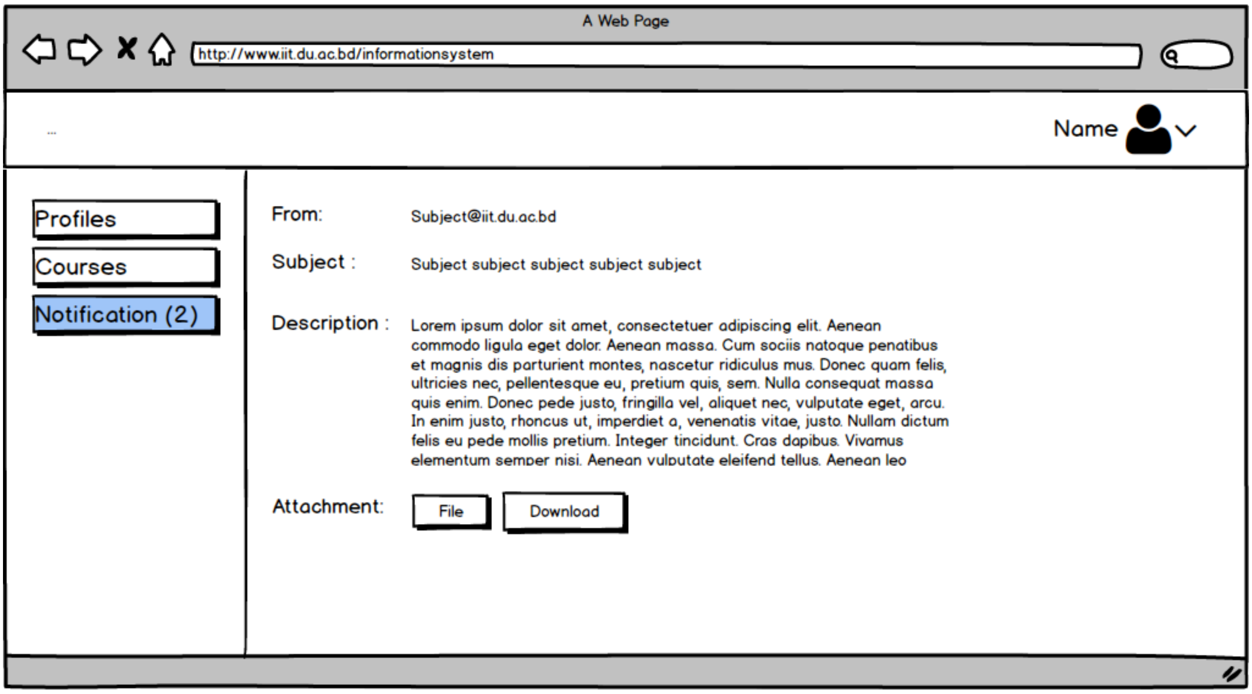

FIGURE 8.15: ADMIN NOTIFICATION 3RD PAGE

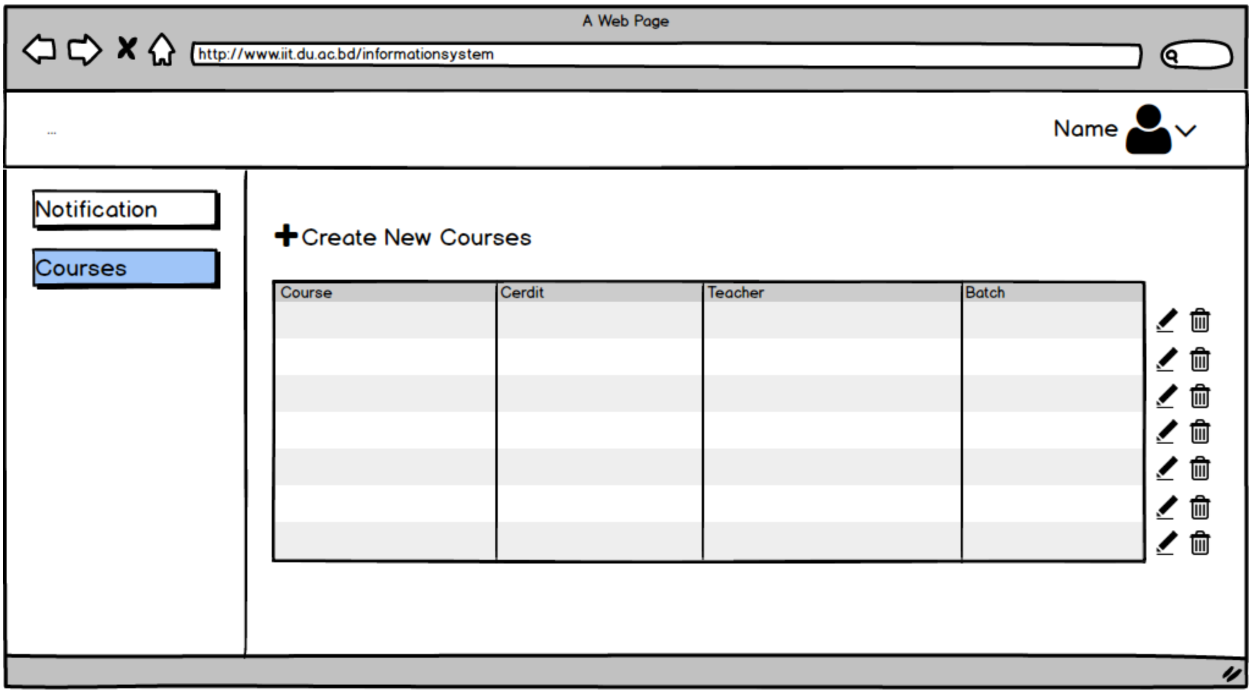

FIGURE 8.16: STAFF COURSE 1ST PAGE

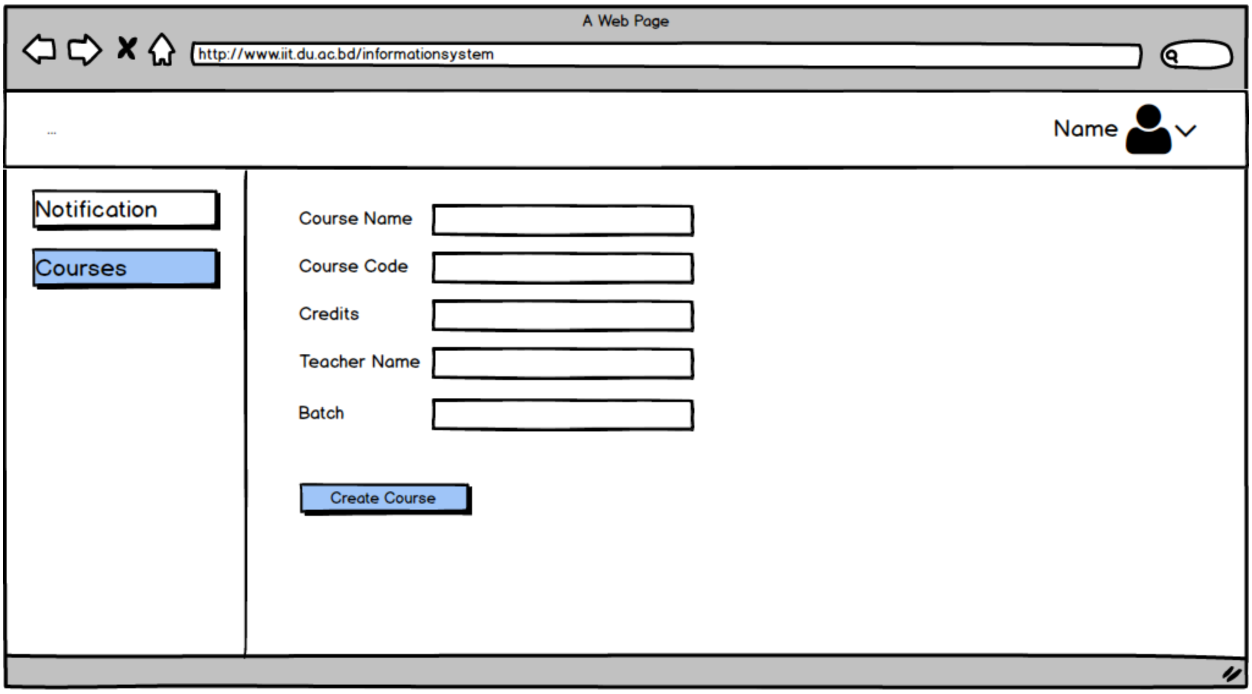

FIGURE 8.17: STAFF COURSE 2ND PAGE

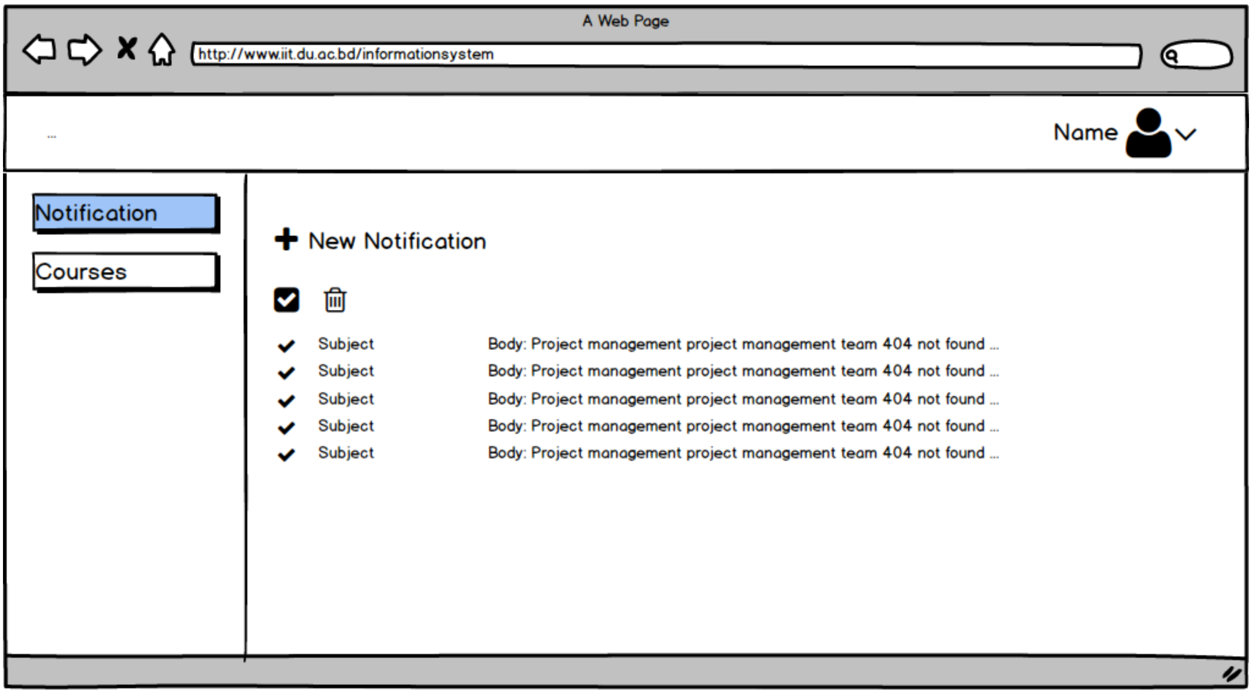

FIGURE 8.18: STAFF NOTIFICATION 1ST PAGE

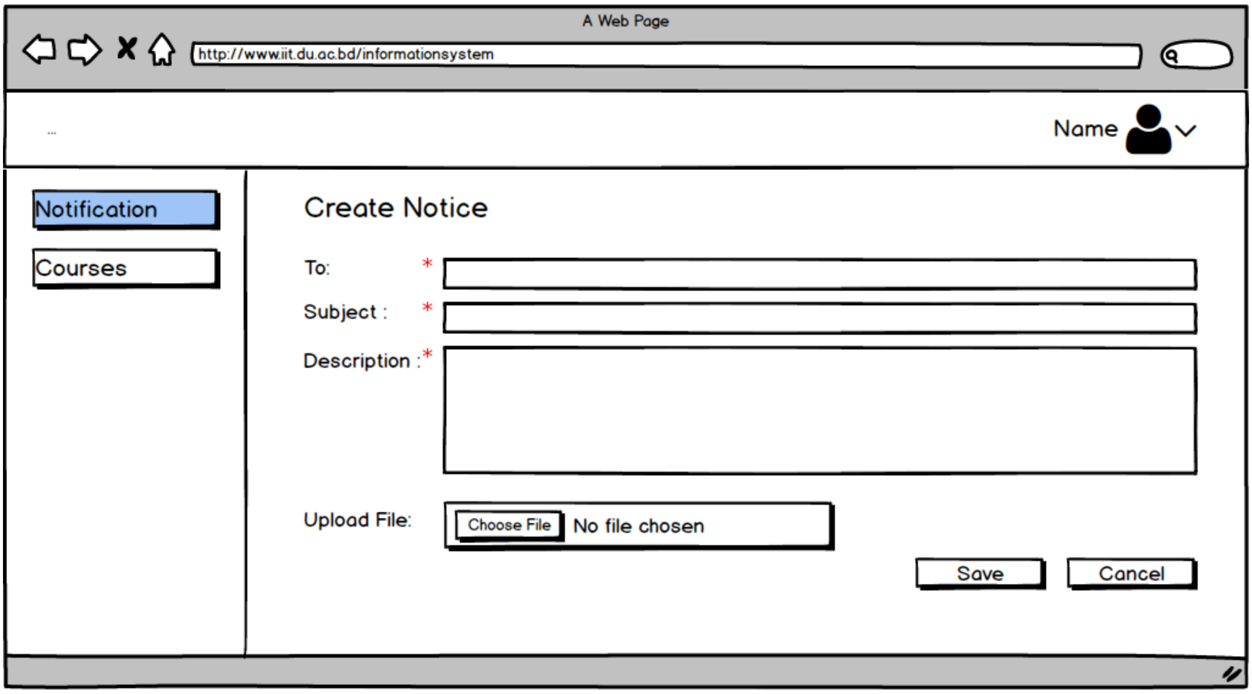

FIGURE 8.19: STAFF NOTIFICATION 2ND PAGE

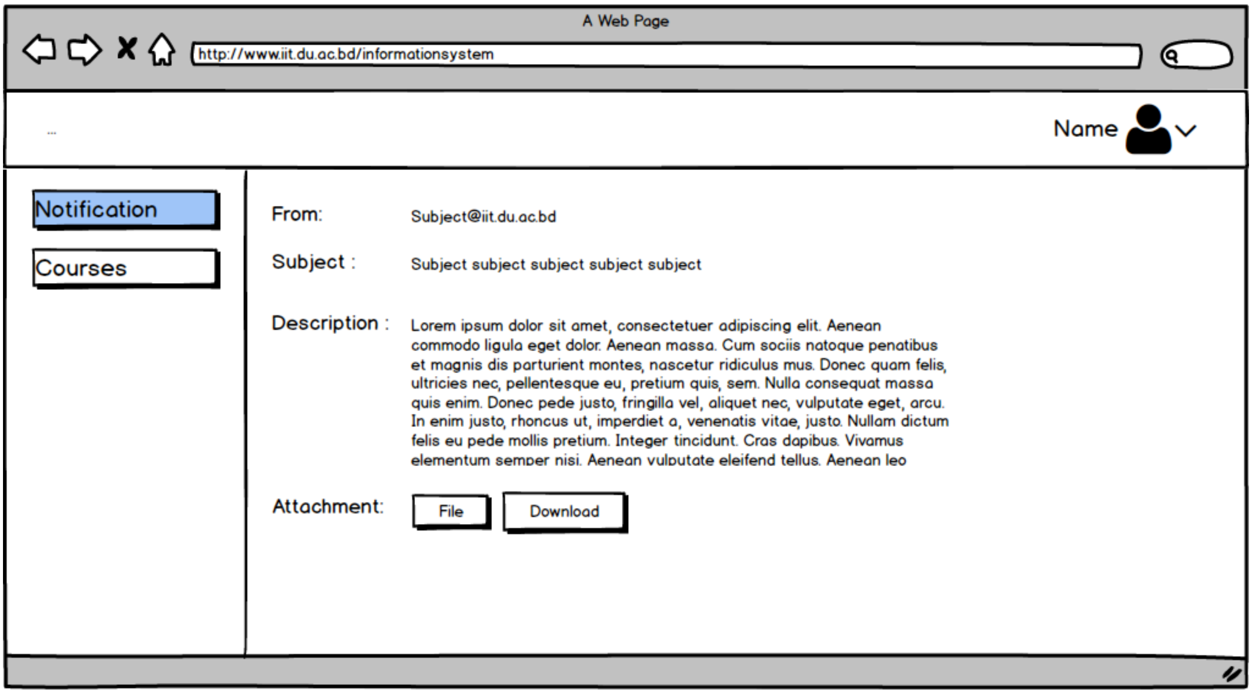

FIGURE 8.20: STAFF NOTIFICATION 3RD PAGE

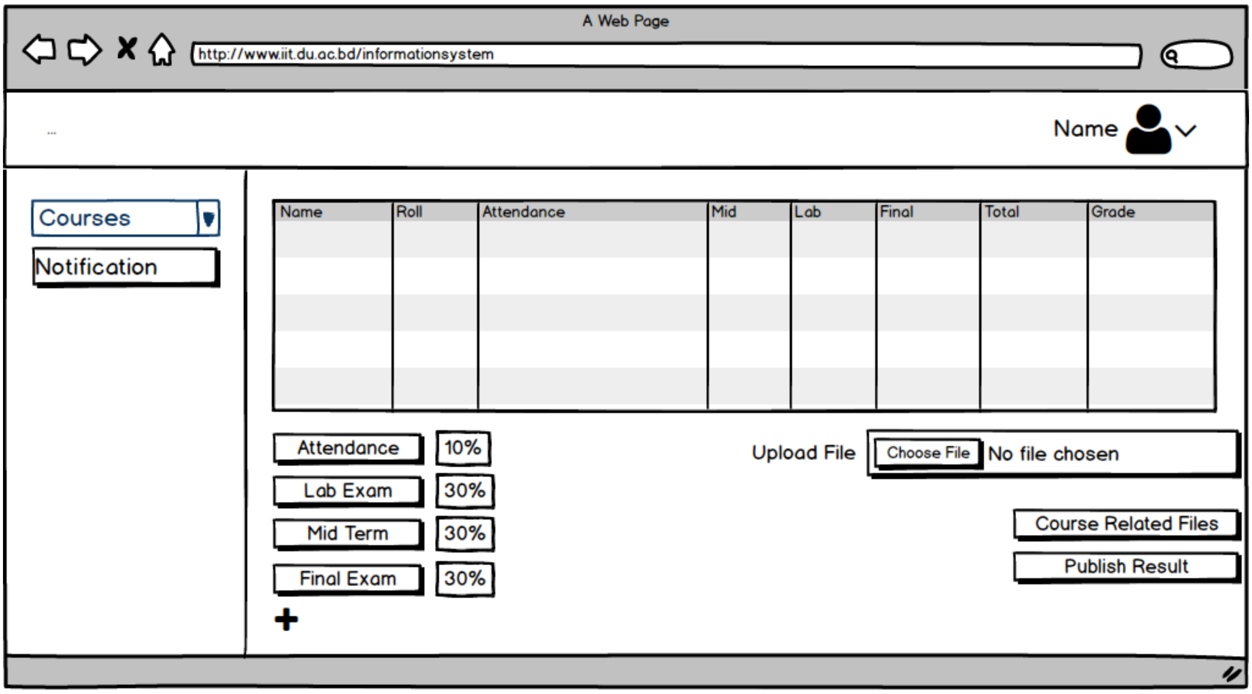

FIGURE 8.21: TEACHER COURSES 1ST PAGE

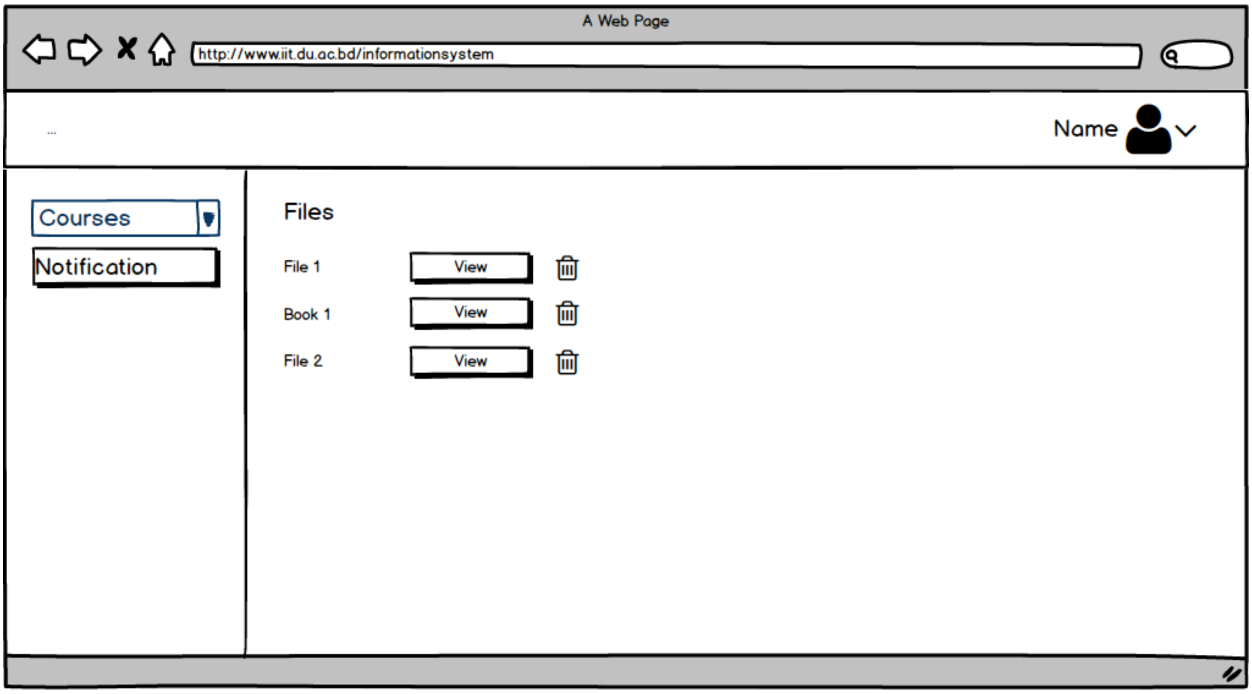

FIGURE 8.22: TEACHER COURSES FILE PAGE

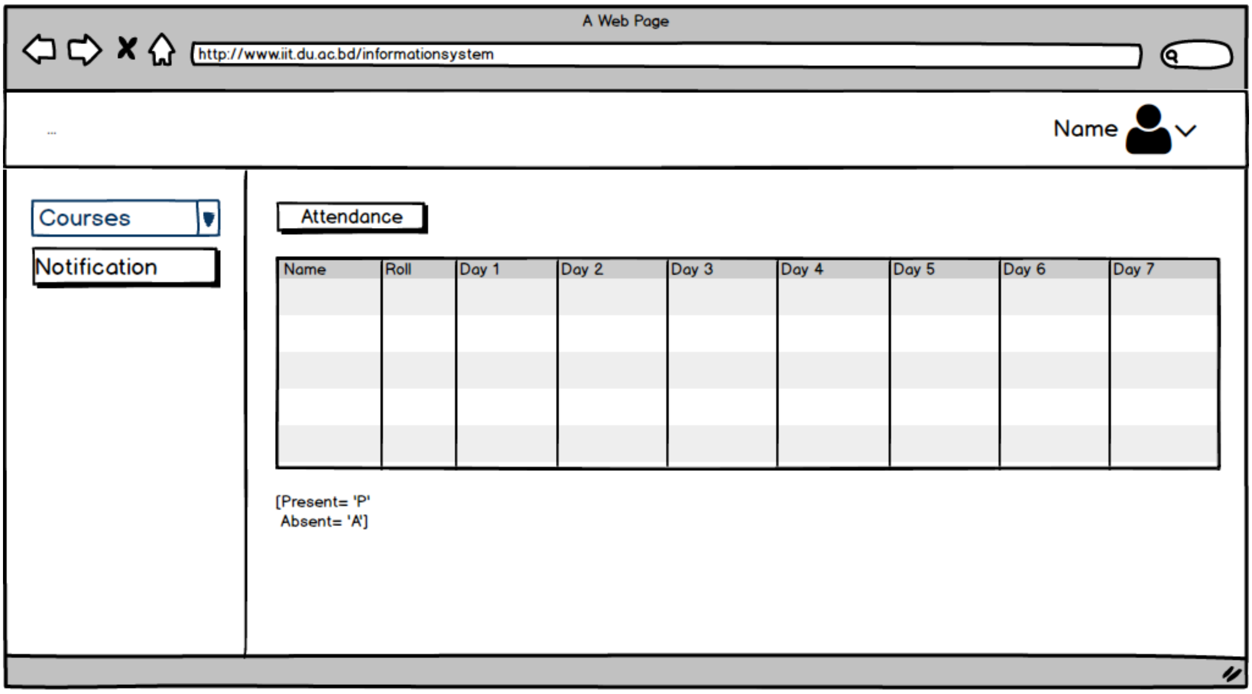

FIGURE 8.23: TEACHER ATTENDANCE PAGE

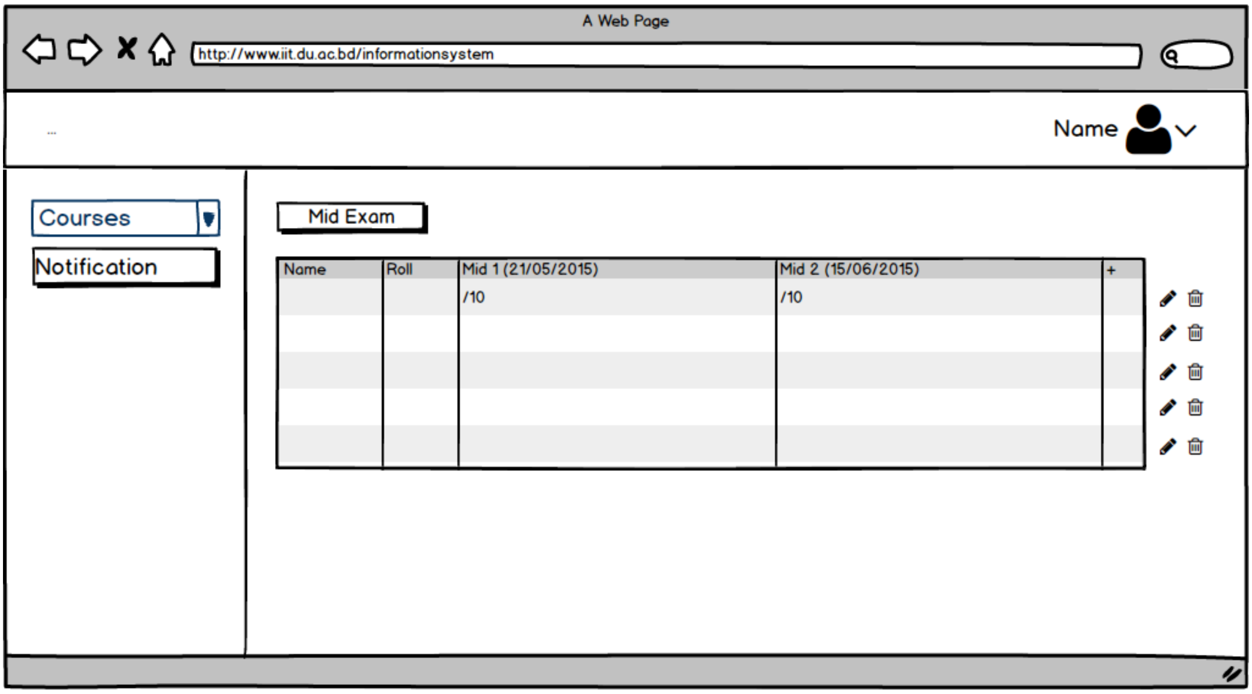

FIGURE 8.24: TEACHER MID PAGE

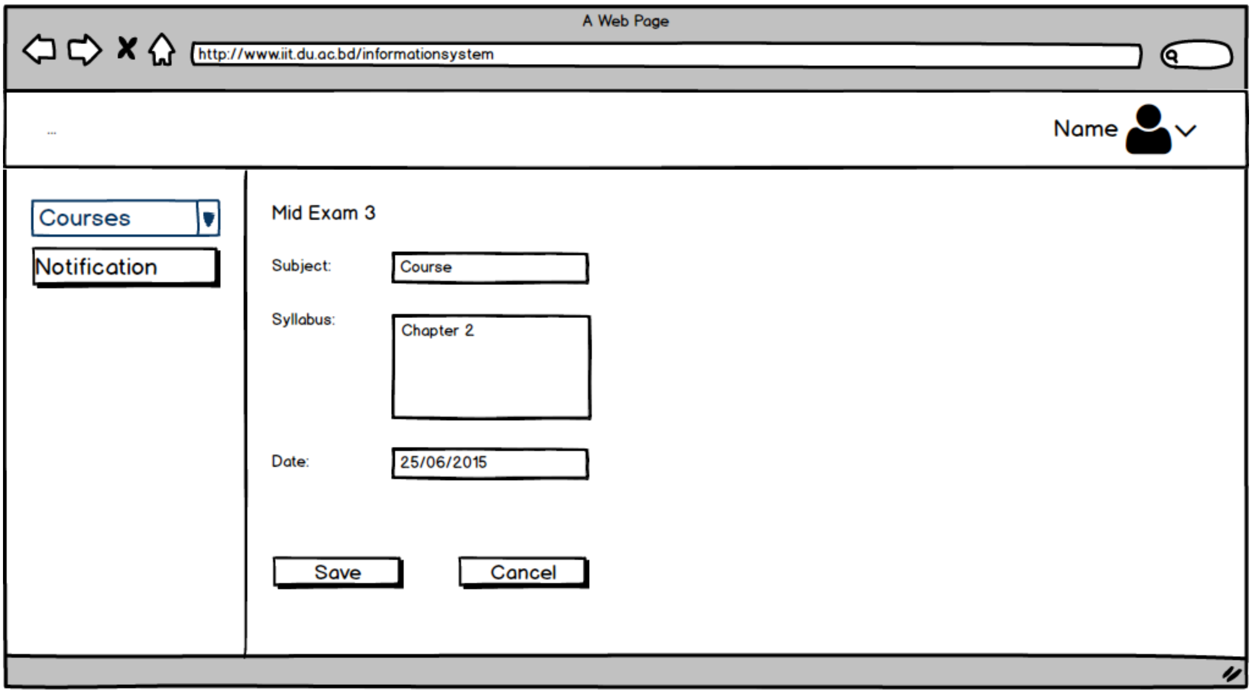

FIGURE 8.25: TEACHER MID 2 PAGE

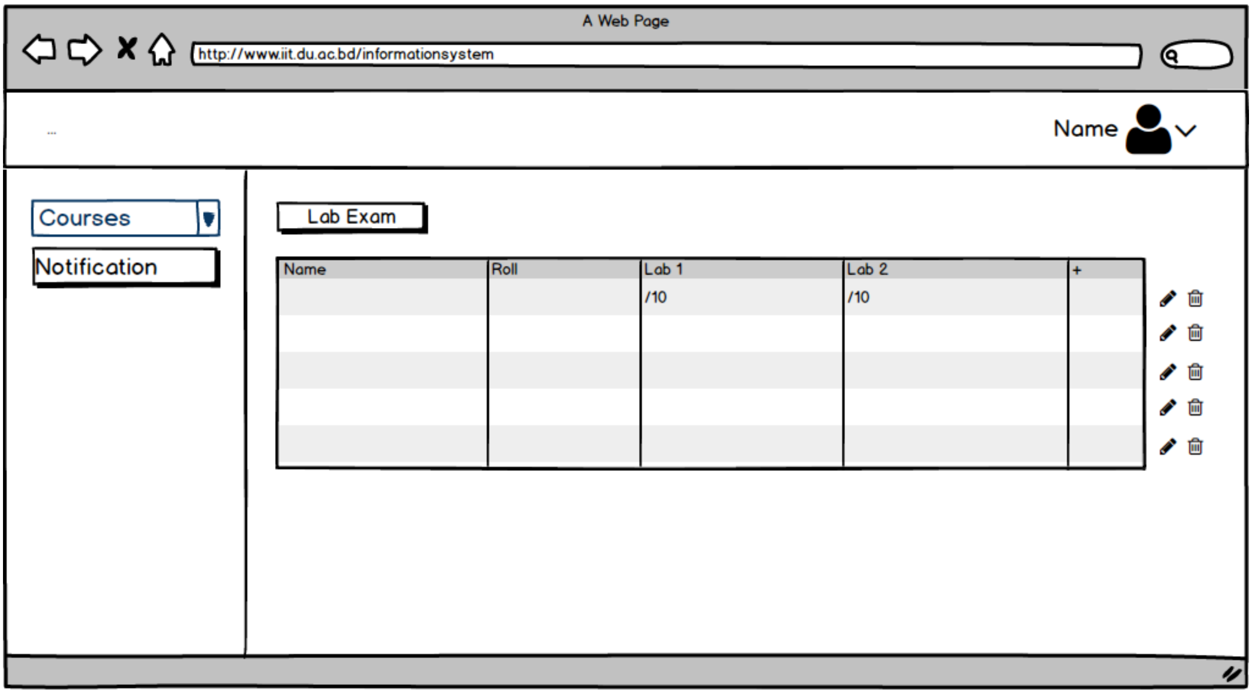

FIGURE 8.26: TEACHER LAB PAGE

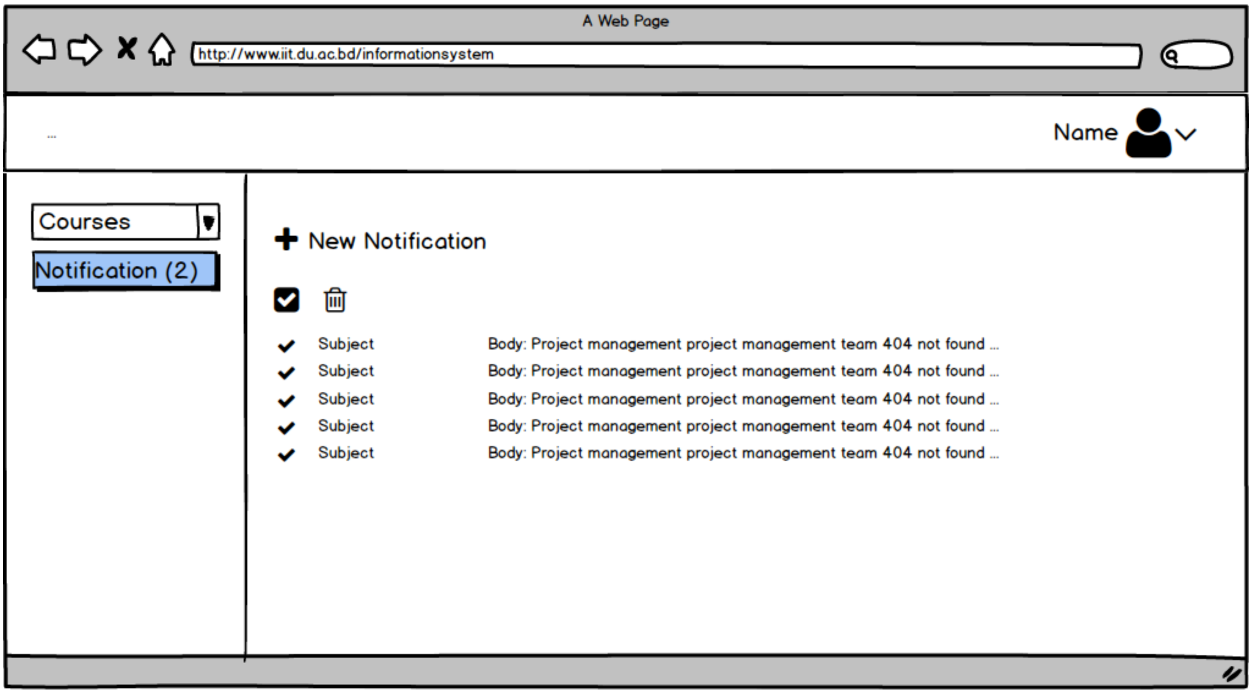

FIGURE 8.27: TEACHER NOTIFICATION 1ST PAGE

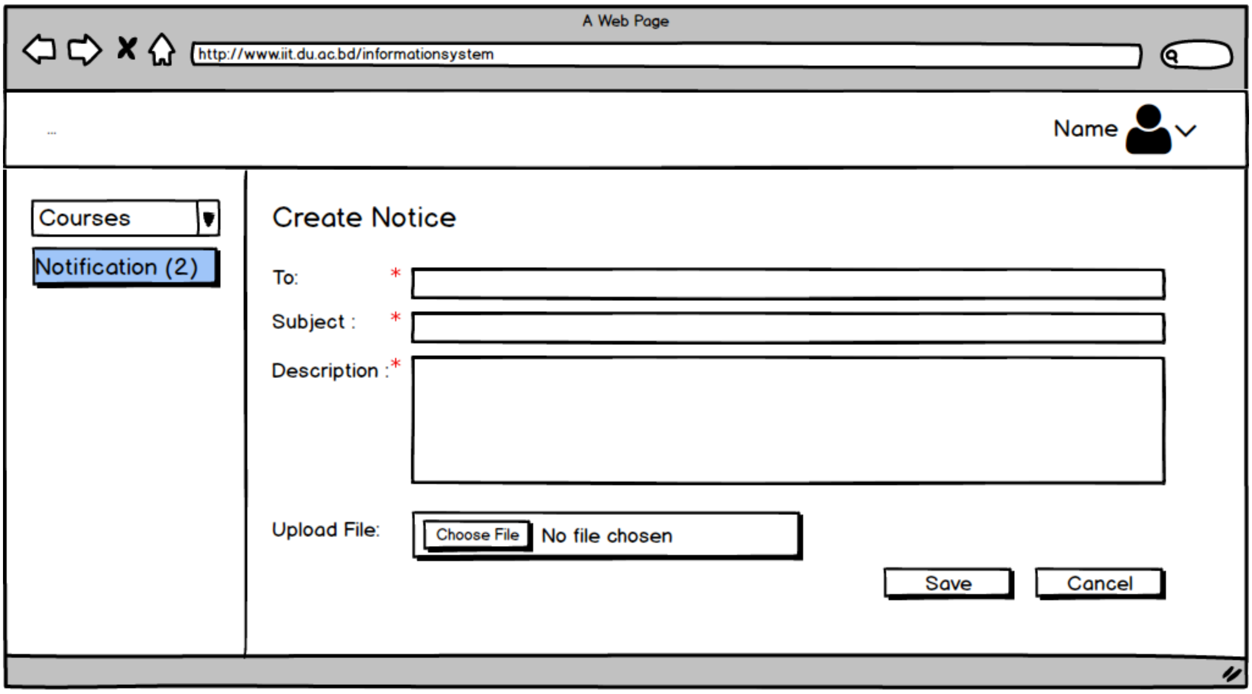

FIGURE 8.28: TEACHER NOTIFICATION 2ND PAGE

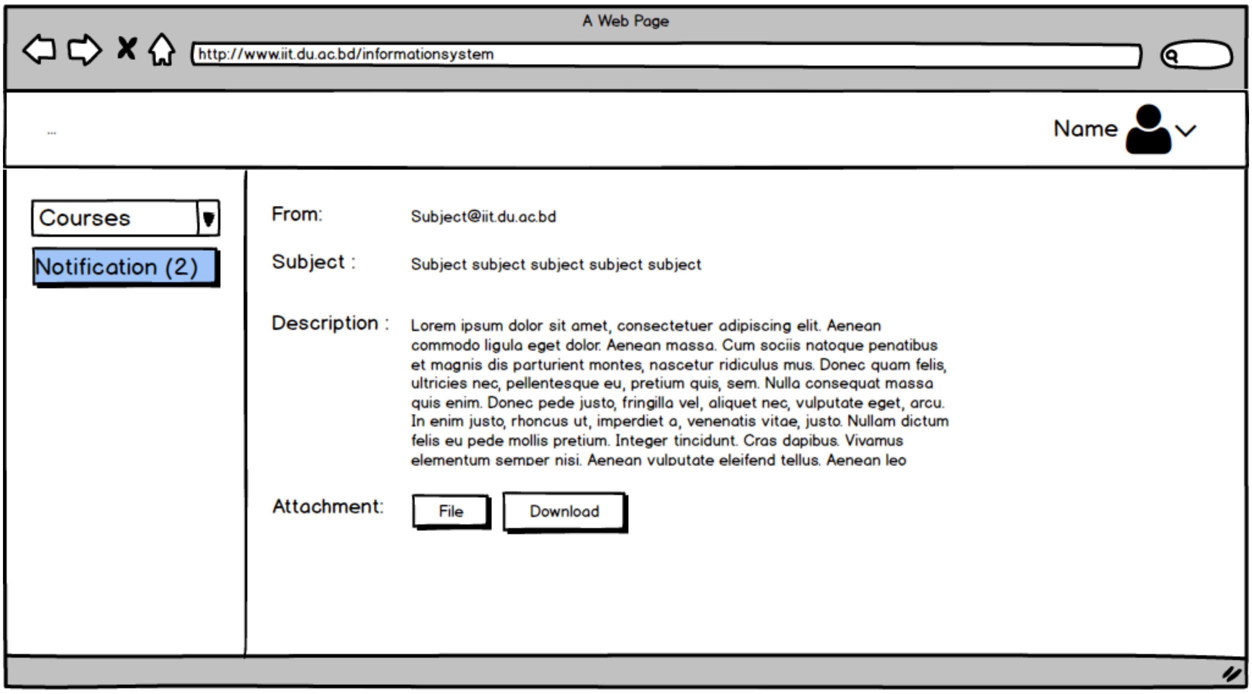

FIGURE 8.29: TEACHER NOTIFICATION 3RD PAGE

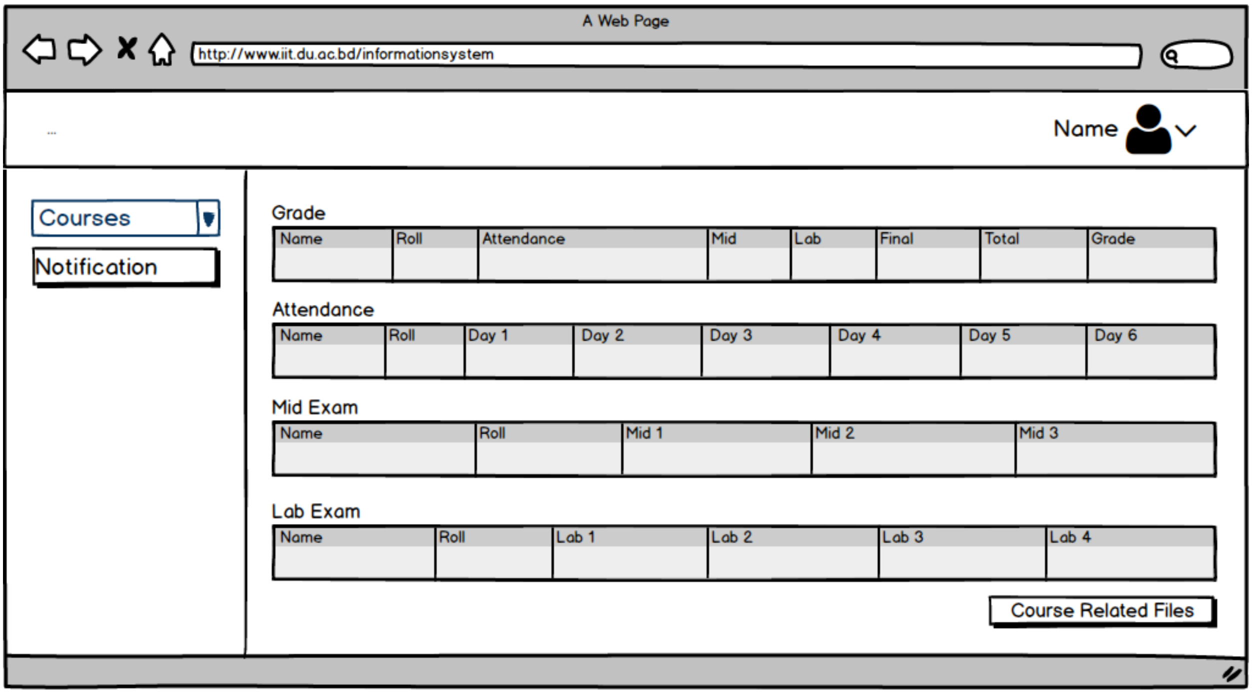

FIGURE 8.30: STUDENT COURSE 1ST PAGE

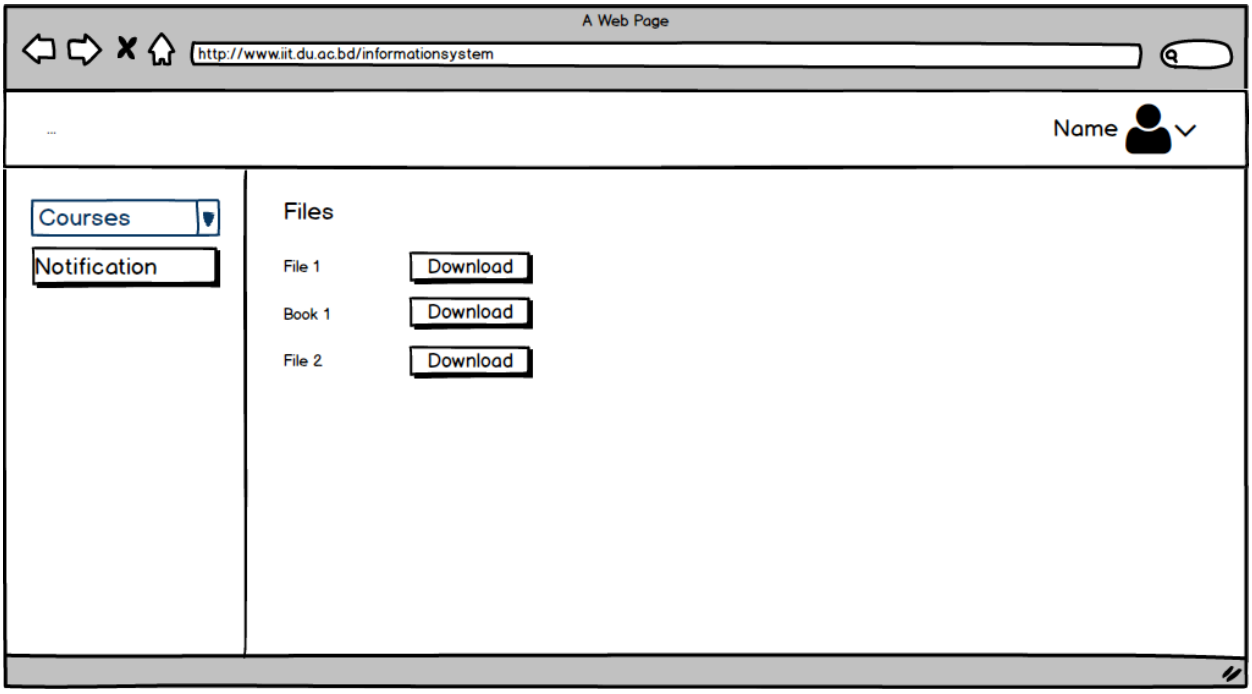

FIGURE 8.31: STUDENT COURSE 2ND PAGE

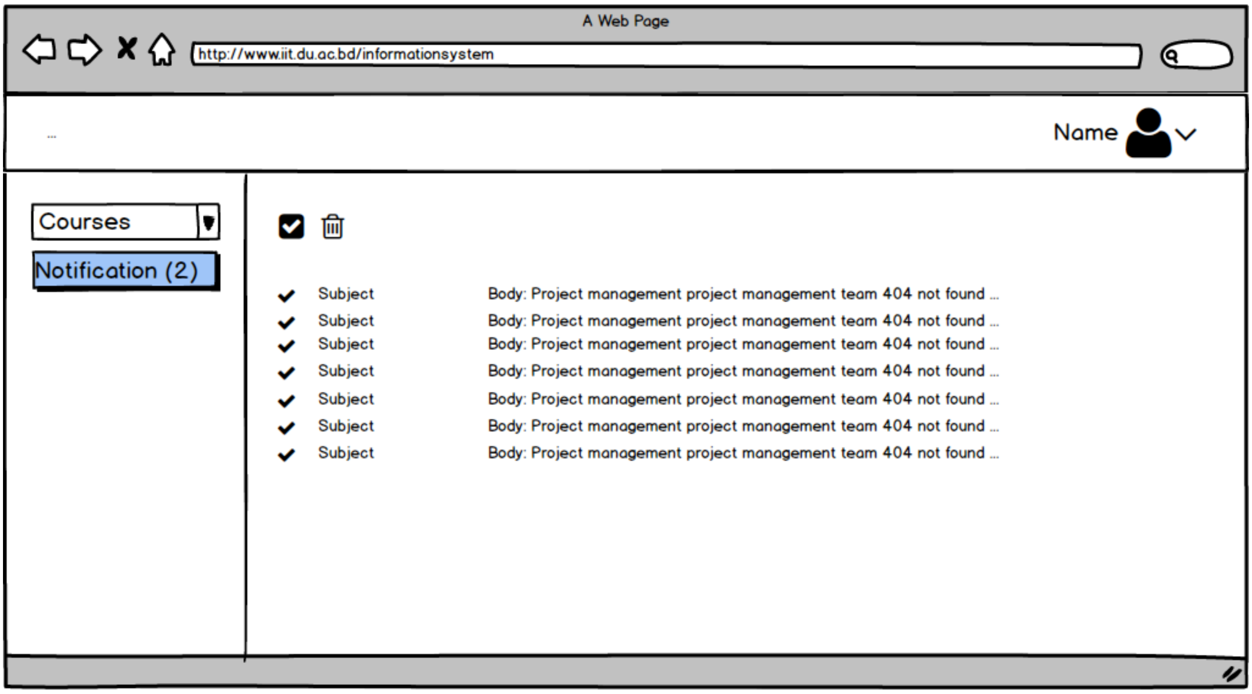

FIGURE 8.32: STUDENT NOTIFICATION 1ST PAGE

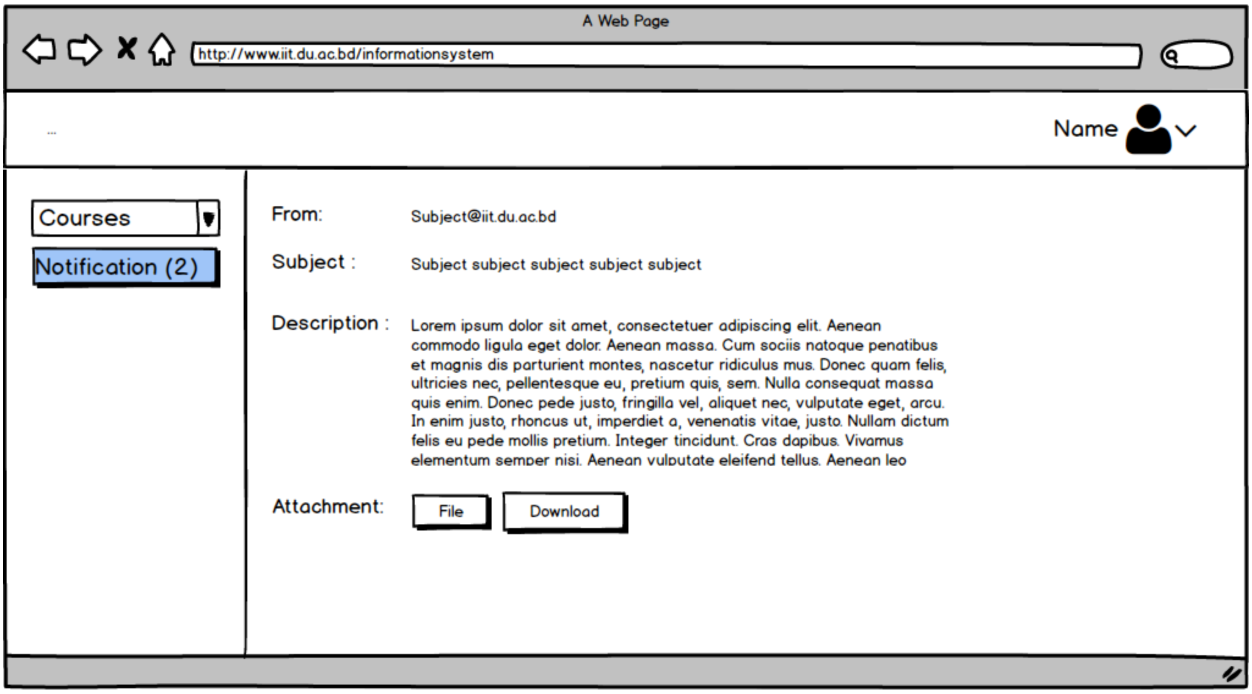

FIGURE 8.33: STUDENT NOTIFICATION 2ND PAGE

CHAPTER 9

CONCLUSION

After a lot of hard working, at last we are able to complete the final SRS report on Student Information Management System. Here we have tried our best to identify necessary requirements from our stakeholders and based on the requirements, we have illustrated different models with diagrams which will help the developers, software designers and other people associated with it to understand about the system and to do their tasks in a better way. Students as well as teachers may use this document as academic learning resource. We hope that the readers will get benefit from the document.